



Sound Attenuators

EXCELAIR

AIR OUTLETS
Technical Catalogue



Sound Attenuators

Sound Attenuators are professionally designed for broad band noise attenuation of fan/impeller noise propagated through all supply and exhaust air systems. Sound attenuators are very sensitive and used in absorbing transmission of high dB sound /noise. We offer our sound attenuators in various shapes and sizes to meet the acoustic engineer's demand.

Ref:GD1000SAT

Sound Attenuators

Introduction

The primary objective for sound attenuation in HVAC systems and equipment is to ensure that the acoustical environment in a given space is maintained. Sound is created by a source, transmitted along one or more paths, and finally reaching a receiver. Treatments and modifications can be applied to any or all of these elements to achieve an acceptable acoustical ambiance, although it is usually most effective and least expensive to attenuate noise at the source.

In HVAC systems, the sound sources are associated with the building, mechanical and electrical equipment. Sound sources are the components that either intrinsically generate noise, like electric motors, or when air passes through them, like dampers or diffusers. Sound receivers are generally the occupant of the building. The noise control engineers are most often constrained to modify the sound transmission paths as a means of achieving the desired sound levels in those areas of a building which are inhibited. In a bid to control noise it is well worth to examine the source of noise which can be fan, duct, diffuser, and building, the sound transmissions path and the receivers.

The noise control/sound attenuation is achieved by means of selective sound absorbing material by absorption course. Silencers or attenuators used in HVAC installation are for treating air borne noise. For better attenuations in mid and high frequency range (0.5 kHz – 8 kHz) sound attenuators are to be in absorptive type design. The acoustic energy propagating through the silencer is effectively absorbed by the fibrous absorption material and converted into heat energy.

The selection shall be on the basis of maximum allowable pressure drop at rated flow and the silencing criteria. Flow area throughout a silencer must be sufficient to accommodate the air flow without imposing excessive restriction. Efforts to controls noise and vibration in a wide range of mechanism and devices inevitably require the use of passive acoustical materials. It requires not only the correct choice of materials but also understanding the properties of the acoustical material how they work,. Also how and where to install them is vital to achieve the greatest performance.

Wide range of sound attenuators are carefully designed and constructed to compact the noise propagation which is generated from various mechanical and electrical equipments used in HVAC system. These sound attenuating products are very sensitive and used in blocking transmission of any unwanted noise. EXCELAIR Sound Attenuators (CSA) are constructed from galvanized steel metal with acoustic insulation in-filled splitters in order to achieve high sound attenuation and reliable performance at all HVAC systems. We offer our sound attenuating devices in various shapes and sizes and they are designed as per acoustic engineers demand.

Rectangular Sound Attenuators are professionally designed for broad band noise attenuation of fan/AHU noises propagating through all exhaust and supply air systems. Its aerodynamically designed bull nosed acoustic splitter provides reliable performance. They are suitable for both low pressure and low frequency applications.

Tubular Sound Attenuators are designed for the noise attenuation of high pressure and high velocity noises propagating through all exhaust and supply air systems. Its aerodynamic design provides high sound attenuation. They are suitable for high pressure, high velocity and low frequency applications. They offer low pressure loss and lowest over all cost.

The Crosstalk sound Attenuators maintain maximum noise suppression and acoustic integrity where privacy and security are essential but regardless of the flow of fresh air. The acoustically engineered aerodynamic design allows maximum and total noise suppression.

In the following catalogue pages, each element of the device is clearly described for quick and accurate understanding of its construction, dimension and performance. Also presented in simple and concise form are the procedures for selecting silencers and calculating the sound attenuation requirements for the overall system.

Sound Attenuators

Terminology

Sound

Sound is defined as a variation in pressure, stress, particle displacement and a velocity in a medium with internal forces. In other words it is, a physical disturbance in a medium (e.g. air) that is capable of being detected by human ear.

Sound Attenuation

Sound attenuation is defined as the reduction of the level of sound as it travels from a source to a receiver. The reduction of sound level per unit distance can be achieved by absorption, diffraction, diffusion, or scattering.

Frequency (HZ)

The frequency of sound is expressed in wavelengths per second (or) cycles per second (CPS). It is more commonly referred to as Hertz. Low frequency noise is 250 Hertz (Hz) and below. High frequency noise is 2 kHz and above. Mid frequency noise falls between 250Hz and 2 kHz. The audible frequency range for humans extends from about 20 Hz to 20 kHz.

Decibel (DB)

The amplitude of sound is expressed in decibels (dB). This is a logarithmic compressed scale dealing in powers of 10 where small increments in dB correspond to large changes in acoustic energy.

Octave Band

An octave band is a frequency band with its upper frequency limit twice as that of its lower frequency limit. Octave band are identified by mid-band frequency, which is the geometric mean of the upper and lower band limits. By international agreements these comprise 63Hz, 125Hz, 250Hz, 500Hz, 1kHz, 2kHz, 4kHz and 8kHz are group of frequencies named by the centre frequency, while the upper frequency is always twice the lower limit of the range.

Sound Pressure Level (LP)

In a given enclosed space (e.g room) this is the noise level which is resonated into a human ear having been radiated from a sound source and subjected to room dispersions and attenuation. Expressed on a decibel scale refers to a reference sound pressure of 2×10^{-5} Pa or 0.0002 microbar.

Sound Power Level (LW)

This represents the total sound energy transmitted by the device (fan, etc.) and is independent of the characteristics of the room into which the sound is radiated. Sound power level cannot be measured directly. It must be calculated from sound pressure level measurements. The sound power level of a source is the ratio expressed in decibels of its sound power divided by the reference sound power which is 10-12 watts.

Sound Intensity

Sound intensity at a point in a specified direction is the rate of flow of sound energy through unit area at that point. Unit of sound intensity is W/m². Sound intensity level (LI) is expressed in dB with a reference quantity of 10-12 W/m².

Regenerated Noise

Medium or high velocity airflow impinging on any obstruction will cause disturbance of the airflow. The resultant turbulence regenerates noise. HVAC duct design components such as elbows, turning vanes, dampers, offsets, take-offs, tees, etc. are sources which regenerate noise. Regenerated noise can be minimized by ensuring smooth airflow conditions.

Sound Attenuators

Terminology

Background Noise

The constant noise level measured, when all the known and active sound sources have been turned off in a building is called background noise.

Ductborne Noise

The noise generated when sound waves are transmitted through the ductworks, both upstream and downstream of a fan.

Insertion Loss

Insertion loss of a silencer element is expressed in dB and is defined as the decrease in sound pressure level or sound intensity level, measured at a fixed receiver location, when the silencer elements is inserted into the path between the source and the receiver. Simply, it is the difference between the two sound levels when measured at the same point before and after the insertion of a silencer. These measurements are typically made in either octave or 1/3 octave bands. Insertion loss is measured in the presence of both forward and reverse flow conditions.

Forward Flow

Forward flow occurs when the airflow in the sound attenuator is in the same direction as the noise propagation.

Reverse Flow

Reverse flow occurs when the airflow in the sound attenuator is in the opposite direction to the noise propagation.

Noise Criteria

Noise criteria are noise levels which are subjectively acceptable in a given environment. The most commonly used criteria are noise criteria curves (NC levels), noise rating curves (NR levels) and dB(A).

Dissipative Silencer

Dissipative silencers use absorptive media such as fiber glass or rockwool as the principal sound absorption mechanism. This type of attenuator is most effective in reducing mid and high frequency sound energy.

Reactive Silencer

Reactive silencers do not use any absorptive media to dissipate sound. They are constructed by tuned resonators or expansion chambers to react and reduce the sound power at designed frequency. They are suitable for applications like hospitals, laboratories or the areas with strict air quality standards.

Active Silencer

Active silencer systems use microphones, loud speakers and appropriate electronic device to reduce in-duct sound by generating inverse-phase sound waves that destructively interfere with the incident sound energy. They can be used to attenuate the both broad band and tonal sound. They are effective in 31.5Hz through 250Hz octave bands and air velocity less than about 7.5 m/s. Active attenuators are not effective to excessive turbulent air flow conditions.

Sound Measurement

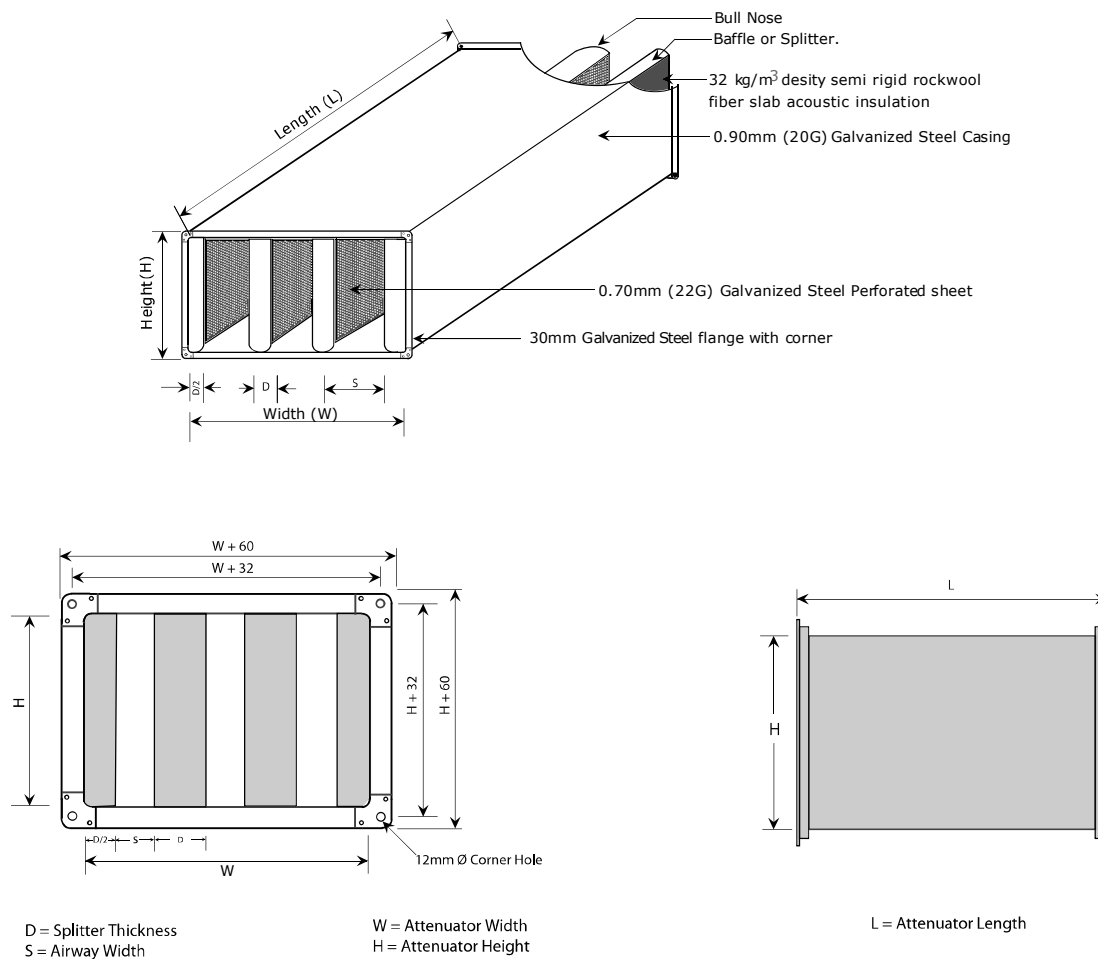
The basic instrument used in measuring sound is sound level meter, which includes a micro phone, electronic circuitry, and a display unit. Sound pressure at a point is converted into sound pressure level and displayed by analogue or digital meters. These devices are usually light, battery operated, hand held units.

Sound Attenuators

Rectangular Sound Attenuator

- Suitable for attenuation of Fan/AHU noises propagated through air ducting
- Highly effective for supply and exhaust air systems.
- Also suitable for cooling towers, air cooled chillers etc.,
- Normal operating temperature range - 10°C to +55°C.
- Suitable for all acoustic, space and energy cost requirements.

Constructional Details (Model: CSA)



Sound Attenuators

Constructional Details

Rectangular sound attenuator casing is constructed from galvanised sheet metal of 0.90mm (20G) thickness. Casing is formed with lock formed seams to meet SMACNA standards. Construction also complies with DW142 class B code and ASTM A924&A653 with 30mm standard Excel flange connection at both ends which is secured to the casing by mechanically closed blind rivets or by pressing. Splitters or baffles are constructed from 0.90mm (20G) thickness of galvanized steel sheet. Splitters are radiussed at both ends to minimize the pressure loss and regenerated noise of the sound attenuator. Splitters are secured to the attenuator casing by means of rivets.

Aerodynamic bull nosed splitters are filled with not less than 32kg/m³ density class 'O' building regulated, inorganic semi rigid Rockwool fiber slab (mineral wool) acoustic filler material. This filler material is inert, non-hygroscopic vermin & rots proof, does not support bacterial & fungus growth. The acoustic filler has a black ceiling veil (black facing) and packed with 0.70mm(22G) thickness galvanized steel perforated sheet, this dual protection prevents damage and fiber erosion up to 30m/s air way velocity.

Sound attenuator with melinex (Polyester membrane) facing is available as option. Polyester membrane is used to protect the acoustic lining from oil or grease or in clean or sterile or hospital applications.

CSA-Sound attenuator model have been tested, rated and certified by Intertek ETL SEMCO America's NVLAP -Accredited laboratories, in accordance with ASTM E477-06 the "Standard Method of Testing Duct Liner Materials and Prefabricated Silencer for Acoustical and Airflow Performance". Test reports can be provided upon request.

The acoustic infill comply the following standards,

Fire resistance	: ASTM E 119 & E84, ASTM C447 & BS 476.
Acoustic Performance	: ASTM C423, ASTM E413 & E90 & BS 3638/ISO 354.
Moisture Absorption	: ASTM C209/14, BS 2972/12, ASTM C1104
Corrosion Resistance	: ASTM C692.

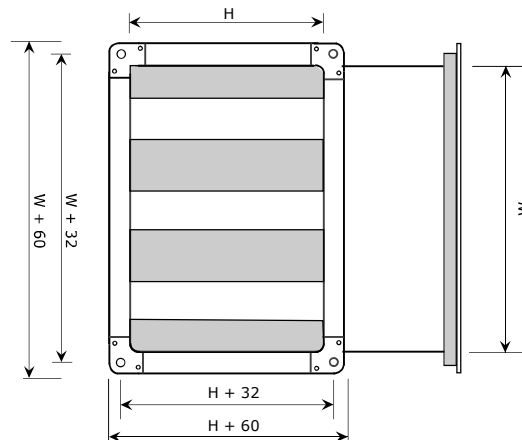
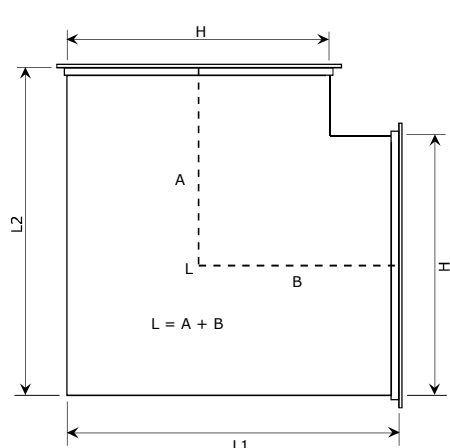
CSA//HP: Sound attenuator model CSA//HP is suitable for high pressure application with 1.2mm (18G) thickness casing with 40mm flange and stiffener to comply the DW144 class C or D and SMACNA standard (high pressure).

CSA//MF: Sound attenuator model CSA//MF is normal rectangular sound attenuator with Melinex facing (Polyester membrane) which is suitable for hospital or oil or grease applications(special applications).

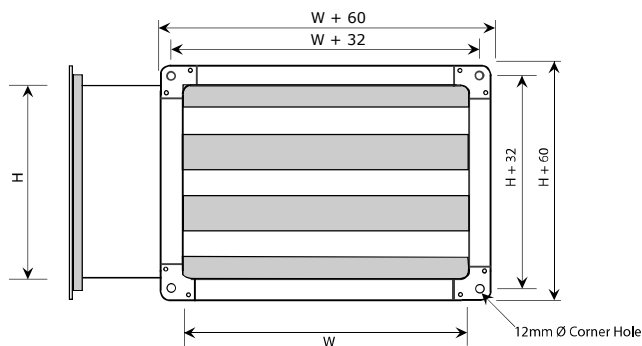
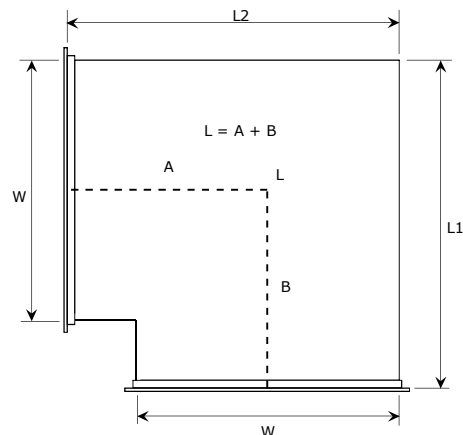
Sound Attenuators

Bend Attenuator

Vertical Bend Attenuator (Model: CSA-BV)



Horizontal Bend Attenuator (Model: CSA-BH)



The construction of bend type attenuator is same as for the straight rectangular attenuators. L denotes ($L = A + B$) the bend centre path length which is equal to the acoustic length (length) mentioned in the selection and performance tables of the normal attenuators. $L1$ and $L2$ are air intake and discharge legs respectively. Unless requested, $L1$ is equal to $L2$ for standard construction.

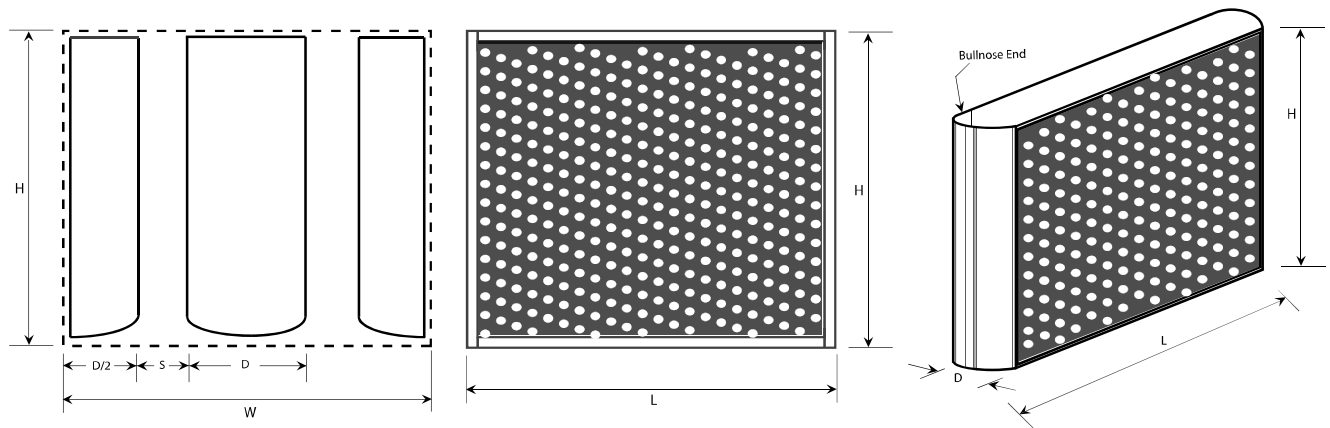
Sound Attenuators

Acoustic Splitter

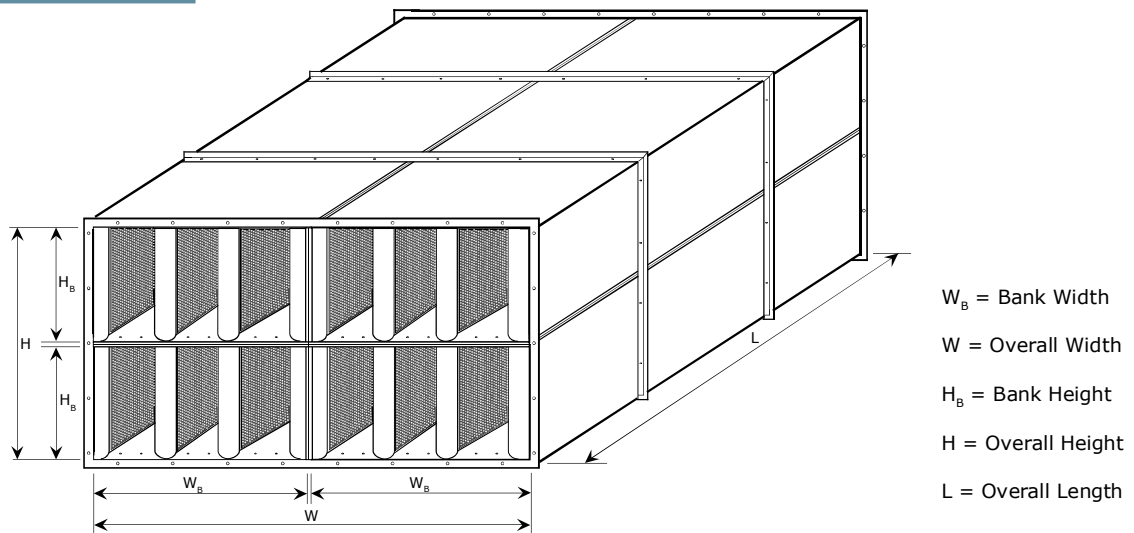
Acoustic Splitter (Model: CAS)

The splitters or baffles are usually filled with acoustic infill, fabricated to fit into builders duct work, various structures, wall openings, shafts and inclusion of AHU sections. It is designed to meet the specified noise level (NC or NR levels) by reducing or attenuating the fan noise.

The construction of acoustic splitter is same as the construction of splitters in straight rectangular attenuators. Aerodynamic bull nosed splitters are filled with not less than 32kg/m^3 density class 'O' building regulated, inorganic semi rigid Rockwool fiber slab (mineral wool) acoustic filler material. The bull nosed construction provides minimum pressure loss and regenerated noise. The combination of acoustic splitter and airway produces an attenuator module.



Multi Section Assembly



Attenuators are supplied in sections when any of the following dimensions are exceeded : Width=2400mm, Height=1800mm and Length=2100mm. Each attenuator can be divided into equal sizes to achieve the actual dimension of the attenuator. The assembly of multi-section (bank) sound attenuators at site to be carried out by the client. The full details of assembly and configuration will be provided by CMS.

Sound Attenuators

Operation and Maintenance

Optional Constructional Specification

Galvanized Steel construction	: 1.0mm GI casing (Model: CSA//HP-1) 1.2mm GI casing (Model: CSA//HP-2)
Stainless Steel Construction	: 1.0mm thickness, Grade 304 with 2B mill finish (Model: CSA//SS-304). 1.0mm thickness, Grade 316 with 2B mill finish (Model: CSA//SS-316). (As per ASTM 240)
Aluminium Construction	: 1.0mm thickness Grade 3003 H14 (Model: CSA//AL-3003)
Additional Coating	: Epoxy coated (model: CSA//EC).
Additional Facing	: Melinex Facing (Model: CSA//MF)
Optional Acoustic Filler	: Fiber Glass (Model: CSA//FG)
Angle Flange	: 30x30x3mm (Model: CSA//AF-30) 40x40x3mm(Model: CSA//AF-40 50x50x5mm(Model: CSA//AF-50)

Operation and Maintenance Instructions

The attenuators should be visually inspected at least once a year to verify that:

- a. Perforated acoustic splitters are undamaged, remaining parallel and rigidly fixed.
- b. Airway widths between the acoustic splitters are free of any debris and blockage.
- c. The holes in the perforated steel are open and free of dust or other foreign matter.
2. In the event that debris must be cleaned from the airway gap or the perforated metal, the attenuator should be vacuum-cleaned or wiped clean with a cloth dampened in a mild detergent solution.
3. Sound attenuators have no moving parts and therefore no lubrication or routine maintenance is required
4. All sound attenuators are furnished rigidly constructed, well-made, and free of any defects in materials or workmanship.
5. In no event should solutions be used to clean attenuators that might affect the zinc coating on the galvanized steel sheets.
6. The occurrence of white rust (Zinc oxide) on galvanized sound attenuators is a normal event and no maintenance is required. It occurs when the zinc in the galvanized steel reacts electrolytically with moisture to protect the steel.
7. Access doors to be provided before and after the sound attenuators for cleaning & maintenance on all the installations.
8. In the event of fire, flood, structural damage or other severe occurrences, contact CMS for specific instructions and recommendations.

Sound Attenuators

Quick Selection

Testing

CSA-Sound attenuators are tested, rated and certified by Intertek ETL SEMCO America's NVLAP –Accredited laboratories, in accordance with ASTM E477-06 the "Standard Method of Testing Duct Liner Materials and Prefabricated Silencer for Acoustical and Airflow Performance".

Sound pressure level data was obtained using a Bruel and Kjaer Digital Frequency Analyzer Type 213. The reference sound source used to obtain the generated sound power level was a calibrated Bruel and Kjaer Digital Frequency Analyzer Type-4204, which conforms to the above standard. Silencer Insertion Loss with and without Airflow and Airflow Generated Sound were recorded.

The Intertek 16,640 cu.ft (470 cubic meters) reverberation room is qualified in accordance with ANSI S 12.32 for sound measurement from 100 to 10,000 hertz. Test reports can be provided upon request.

Quick Selection

Attenuator Face Velocity Vs NC level Required:

Attenuator Model	Maximum permissible face velocity V_f in m/s against NC level required					
	NC-25	NC-30	NC-35	NC-40	NC-45	NC-50
CSA-H/20-75	2.5	3.0	4.0	4.5	5.5	6.3
CSA-HM/20-100	3.0	4.0	5.0	6.2	7.0	7.5
CSA-M/20-125	3.5	5.0	6.0	7.0	8.0	8.5
CSA-MA/20-150	4.0	5.5	6.5	7.5	9.0	10.0
CSA-LM/20-175	4.5	6.5	8.0	9.0	10.0	10.5
CSA-L/20-200	5.0	7.0	8.5	9.5	10.5	11.5

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-H/20-75

Pressure Drop ΔP (Pa)				18	27	35	48	62	75	86	104	115	132
Face Velocity V _f (m/s)				1.5	2.0	2.5	2.8	3.3	3.5	3.8	4.0	4.3	4.6
Self generated noise level against face velocity V _f				NC 25			NC 30		NC 35			NC 40	
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)									
Width (mm)	Height (mm)												
275	150	0.04	ONE	62	83	103	112	136	144	157	165	177	190
	300	0.08		124	165	206	224	272	289	314	330	335	380
	450	0.12		186	248	309	336	408	433	470	495	532	569
	600	0.17		248	330	413	476	545	578	627	660	710	759
	750	0.21		309	413	516	588	681	722	784	825	887	949
	900	0.25		371	495	619	700	817	866	941	990	1064	1139
550	300	0.17	TWO	248	330	413	476	545	578	627	660	710	759
	450	0.25		371	495	619	700	817	866	941	990	1064	1139
	600	0.33		495	660	825	924	1089	1155	1254	1320	1419	1518
	750	0.41		619	825	1031	1148	1361	1444	1568	1650	1774	1898
	900	0.50		743	990	1238	1400	1634	1733	1881	1980	2129	2277
	1050	0.58		866	1155	1444	1624	1906	2021	2195	2310	2483	2657
825	450	0.37	THREE	557	743	928	1036	1225	1299	1411	1485	1596	1708
	600	0.50		743	990	1238	1400	1634	1733	1881	1980	2129	2277
	750	0.62		928	1238	1547	1736	2042	2166	2351	2475	2661	2846
	900	0.74		1114	1485	1856	2072	2450	2599	2822	2970	3193	3416
	1050	0.87		1299	1733	2166	2436	2859	3032	3292	3465	3725	3985
	1200	0.99		1485	1980	2475	2772	3267	3465	3762	3960	4257	4554
1100	600	0.66	FOUR	990	1320	1650	1848	2178	2310	2508	2640	2838	3036
	750	0.83		1238	1650	2063	2324	2723	2888	3135	3300	3548	3795
	900	0.99		1485	1980	2475	2772	3267	3465	3762	3960	4257	4554
	1050	1.16		1733	2310	2888	3248	3812	4043	4389	4620	4967	5353
	1200	1.32		1980	2640	3300	3696	4356	4620	5016	5280	5676	6072
	1350	1.49		2228	2970	3713	4172	4901	5198	5643	5940	6386	6831
1375	750	1.03	FIVE	1547	2063	2578	2884	3403	3609	3919	4125	4434	4744
	900	1.24		1856	2475	3094	3472	4084	4331	4703	4950	5321	5693
	1050	1.44		2166	2888	3609	4032	4764	5053	5486	5775	6208	6641
	1200	1.65		2475	3300	4125	4620	5445	5775	6270	6600	7095	7590
	1350	1.86		2784	3713	4641	5208	6126	6497	7054	7425	7982	8539
	1500	2.06		3094	4125	5156	5768	6806	7219	7838	8250	8869	9488
1650	900	1.49	SIX	2228	2970	3713	4172	4901	5198	5643	5940	6386	6831
	1050	1.73		2599	3465	4331	4844	5717	6064	6584	6930	7450	7970
	1200	1.98		2970	3960	4950	5544	6534	6930	7524	7920	8514	9108
	1350	2.23		3341	4455	5569	6244	7351	7796	8465	8910	9578	10247
	1500	2.48		3713	4950	6188	6944	8168	8663	9405	9900	10643	11385
	1650	2.72		4084	5445	6806	7616	8984	9529	10346	10890	11707	12524
1925	1800	2.97	SEVEN	4455	5940	7425	8316	9801	10395	11286	11880	12771	13662
	1050	2.02		3032	4043	5053	5656	6670	7074	7681	8085	8691	9298
	1200	2.31		3465	4620	5775	6468	7623	8085	8778	9240	9933	10626
	1350	2.60		3898	5198	6497	7280	8576	9096	9875	10395	11175	11954
	1500	2.89		4331	5775	7219	8092	9529	10106	10973	11550	12416	13283
	1650	3.18		4764	6353	7941	8904	10482	11117	12070	12705	13658	14611
2200	1800	3.47	EIGHT	5198	6930	8663	9716	11435	12128	13167	13860	14900	15939
	1950	3.75		5631	7508	9384	10500	12387	13138	14264	15015	16141	17267
	2100	4.04		6064	8085	10106	11312	13340	14149	15362	16170	17383	18596
	1200	2.64		3960	5280	6600	7392	8712	9240	10032	10560	11352	12144
	1350	2.97		4455	5940	7425	8316	9801	10395	11286	11880	12771	13662
	1500	3.30		4950	6600	8250	9240	10890	11550	12540	13200	14190	15180
2475	1650	3.63	NINE	5445	7260	9075	10164	11979	12705	13794	14520	15609	16698
	1800	3.96		5940	7920	9900	11088	13068	13860	15048	15840	17028	18216
	1950	4.29		6435	8580	10725	12012	14157	15015	16302	17160	18447	19734
	2100	4.62		6930	9240	11550	12936	15246	16170	17556	18480	19866	21252
	2250	4.95		7425	9900	12375	13860	16335	17325	18810	19800	21285	22770
	1350	3.34		5012	6683	8353	9352	11026	11694	12697	13365	14367	15370
275	1500	3.71	ONE	5569	7425	9281	10388	12251	12994	14108	14850	15964	17078
	1650	4.08		6126	8168	10209	11424	13476	14293	15518	16335	17560	18785
	1800	4.46		6683	8910	11138	12488	14702	15593	16929	17820	19157	20493
	1950	4.83		7239	9653	12066	13524	15927	16892	18340	19305	20753	22201
	2100	5.20		7796	10395	12994	14560	17152	18191	19751	20790	22349	23909
	2250	5.57		8353	11138	13922	15596	18377	19491	21161	22275	23946	25616
275	2400	5.94	TWO	8910	11880	14850	16632	19602	20790	22572	23760	25542	27324

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths. Pressure loss data assumes that the airflow to the attenuator is uniform over the face in a duct to duct layout.

Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.89	X 0.96	X 1.00	X 1.06	X 1.10	X 1.16	X 1.21

For bend type attenuators 35% to be added to the pressure drop of the straight attenuator.

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-HM/20-100

Pressure Drop ΔP (Pa)				18	27	35	48	62	75	86	104	115	132	
Face Velocity V _f (m/s)				2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.2	
Self generated noise level against face velocity V _f				NC 25			NC 30			NC 35			NC 40	
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)										
Width (mm)	Height (mm)													
300	150	0.05	ONE	90	113	135	158	180	203	225	248	270	293	
	300	0.09		180	225	270	315	360	405	450	495	540	585	
	450	0.14		270	338	405	473	540	608	675	743	810	878	
	600	0.18		360	450	540	630	720	810	900	990	1180	1170	
	750	0.23		450	563	675	788	900	1013	1125	1238	1350	1463	
	900	0.27		540	675	810	945	1080	1215	1350	1485	1620	1755	
600	300	0.18	TWO	360	450	540	630	720	810	900	990	1080	1170	
	450	0.27		540	675	810	945	1080	1215	1350	1485	1620	1755	
	600	0.36		720	900	1080	1260	1440	1620	1800	1980	2160	2340	
	750	0.45		900	1125	1350	1575	1800	2025	2250	2475	2700	2925	
	900	0.54		1080	1350	1620	1890	2160	2430	2700	2970	3240	3510	
	1050	0.63		1260	1575	1890	2205	2520	2835	3150	3465	3780	4095	
900	1200	0.72	THREE	1440	1800	2160	2520	2880	3240	3600	3960	4320	4680	
	450	0.41		810	1013	1215	1418	1620	1823	2025	2228	2430	2633	
	600	0.54		1080	1350	1620	1890	2160	2430	2700	2970	3240	3510	
	750	0.68		1350	1688	2025	2363	2700	3038	3375	3713	4050	4388	
	900	0.81		1620	2025	2430	2835	3240	3645	4050	4455	4860	5265	
	1050	0.95		1890	2363	2835	3308	3780	4253	4725	5198	5670	6143	
1200	1200	1.08	FOUR	2160	2700	3240	3780	4320	4860	5400	5940	6480	7020	
	1350	1.22		2430	3038	3645	4253	4860	5468	6075	6683	7290	7898	
	600	0.72		1440	1800	2160	2520	2880	3240	3600	3960	4320	4680	
	750	0.90		1800	2250	2700	3150	3600	4050	4500	4950	5400	5850	
	900	1.08		2160	2700	3240	3780	4320	4860	5400	5940	6480	7020	
	1050	1.26		2520	3150	3780	4410	5040	5670	6300	6930	7560	8190	
1500	1200	1.44	FIVE	2880	3600	4320	5040	5760	6480	7200	7920	8640	9360	
	1350	1.62		3240	4050	4860	5670	6480	7290	8100	8910	9720	10530	
	1500	1.80		3600	4500	5400	6300	7200	8100	9000	9900	10800	11700	
	750	1.13		2250	2813	3375	3938	4500	5063	5625	6188	6750	7313	
	900	1.35		2700	3375	4050	4725	5400	6075	6750	7425	8100	8775	
	1050	1.58		3150	3938	4725	5513	6300	7088	7875	8663	9450	10238	
1800	1200	1.80	SIX	3600	4500	5400	6300	7200	8100	9000	9900	10800	11700	
	1350	2.03		4050	5063	6075	7088	8100	9113	10125	11138	12150	13163	
	1500	2.25		4500	5625	6750	7875	9000	10125	11250	12375	13500	14625	
	1650	2.48		4950	6188	7425	8663	9900	11138	12375	13613	14850	16088	
	1800	2.70		5400	6750	8100	9450	10800	12150	13500	14850	16200	17550	
	900	1.62		3240	4050	4860	5670	6480	7290	8100	8910	9720	10530	
2100	1050	1.89	SEVEN	3780	4725	5670	6615	7560	8505	9450	10395	11340	12285	
	1200	2.16		4320	5400	6480	7560	8640	9720	10800	11880	12960	14040	
	1350	2.43		4860	6075	7290	8505	9720	10935	12150	13365	14580	15795	
	1500	2.70		5400	6750	8100	9450	10800	12150	13500	14850	16200	17550	
	1650	2.97		5940	7425	8910	10395	11880	13365	14850	16335	17820	19305	
	1800	3.24		6480	8100	9720	11340	12960	14580	16200	17820	19440	21060	
2400	1950	3.51	EIGHT	7020	8775	10530	12285	14040	15795	17550	19305	21060	22815	
	1050	2.21		4410	5513	6615	7718	8820	9923	11025	12128	13230	14333	
	1200	2.52		5040	6300	7560	8820	10080	11340	12600	13860	15120	16380	
	1350	2.84		5670	7088	8505	9923	11340	12758	14175	15593	17010	18428	
	1500	3.15		6300	7875	9450	11025	12600	14175	15750	17325	18900	20475	
	1650	3.47		6930	8663	10395	12128	13860	15593	17325	19058	20790	22523	
2400	1800	3.78	EIGHT	7560	9450	11340	13230	15120	17010	18900	20790	22680	24570	
	1950	4.10		8190	10238	12285	14333	16380	18428	20475	22523	24570	26618	
	2100	4.41		8820	11025	13230	15435	17640	19845	22050	24255	26460	28665	
	2250	4.73		9450	11813	14175	16538	18900	21263	23625	25988	28350	30713	
	1200	2.88		5760	7200	8640	10080	11520	12960	14400	15840	17280	18720	
	1350	3.24		6480	8100	9720	11340	12960	14580	16200	17820	19440	21060	
2400	1500	3.60	EIGHT	7200	9000	10800	12600	14400	16200	18000	19800	21600	23400	
	1650	3.96		7920	9900	11880	13860	15840	17820	19800	21780	23760	25740	
	1800	4.32		8640	10800	12960	15120	17280	19440	21600	23760	25920	28080	
	1950	4.68		9360	11700	14040	16380	18720	21060	23400	25740	28080	30420	
	2100	5.04		10080	12600	15120	17640	20160	22680	25200	27720	30240	32760	
	2250	5.40		10800	13500	16200	18900	21600	24300	27000	29700	32400	35100	
2400	2400	5.76	EIGHT	11520	14400	17280	20160	23040	25920	28800	31680	34560	37440	

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths,

Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.90	X 0.95	X 1.00	X 1.05	X 1.11	X 1.14	X 1.19

For bend type attenuators 40% to be added to the pressure drop of the straight attenuator.

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-M/20-125

Pressure Drop ΔP (Pa)			18	27	35	48	62	75	86	104	115	132	
Face Velocity V _f (m/s)			2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	
Self generated noise level against face velocity V _f			NC 25			NC 30			NC 35		NC 40		
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)									
Width (mm)	Height (mm)												
325	150	0.05	ONE	122	146	171	195	219	244	268	293	317	366
	300	0.10		244	293	341	390	439	488	536	585	634	731
	450	0.15		366	439	512	585	658	731	804	878	951	1097
	600	0.20		488	585	683	780	878	975	1073	1170	1268	1463
	750	0.24		609	731	853	975	1097	1219	1341	1463	1584	1828
	900	0.29		731	878	1024	1170	1316	1463	1609	1755	1901	2194
650	450	0.29	TWO	731	878	1024	1170	1316	1463	1609	1755	1901	2194
	600	0.39		975	1170	1365	1560	1755	1950	2145	2340	2535	2925
	750	0.49		1219	1463	1706	1950	2194	2438	2681	2925	3169	3656
	900	0.59		1463	1755	2048	2340	2633	2925	3218	3510	3803	4388
	1050	0.68		1706	2048	2389	2730	3071	3413	3754	4095	4436	5119
	1200	0.78		1950	2340	2730	3120	3510	3900	4290	4680	5070	5850
975	600	0.59	THREE	1463	1755	2048	2340	2633	2925	3218	3510	3803	4388
	750	0.73		1828	2194	2559	2925	3291	3656	4022	4388	4753	5484
	900	0.88		2194	2633	3071	3510	3949	4388	4826	5265	5704	6581
	1050	1.02		2559	3071	3583	4095	4607	5119	5631	6143	6654	7678
	1200	1.17		2925	3510	4095	4680	5265	5850	6435	7020	7605	8775
	1350	1.32		3291	3949	4607	5265	5923	6581	7239	7898	8556	9872
1300	1500	1.46	FOUR	3656	4388	5119	5850	6581	7313	8044	8775	9506	10969
	750	0.98		2438	2925	3413	3900	4388	4875	5363	5850	6338	7313
	900	1.17		2925	3510	4095	4680	5265	5850	6435	7020	7605	8775
	1050	1.37		3413	4095	4778	5460	6143	6825	7508	8190	8873	10238
	1200	1.56		3900	4680	5460	6240	7020	7800	8580	9360	10140	11700
	1350	1.76		4388	5265	6143	7020	7898	8775	9653	10530	11408	13163
1625	1500	1.95	FIVE	4875	5850	6825	7800	8775	9750	10725	11700	12675	14625
	1650	2.15		5363	6435	7508	8580	9653	10725	11798	12870	13943	16088
	1800	2.34		5850	7020	8190	9360	10530	11700	12870	14040	15210	17550
	900	1.46		3656	4388	5119	5850	6581	7313	8044	8775	9506	10969
	1050	1.71		4266	5119	5972	6825	7678	8531	9384	10238	11091	12797
	1200	1.95		4875	5850	6825	7800	8775	9750	10725	11700	12675	14625
1950	1350	2.19	SIX	5484	6581	7678	8775	9872	10969	12066	13163	14259	16453
	1500	2.44		6094	7313	8531	9750	10969	12188	13406	14625	15844	18281
	1650	2.68		6703	8044	9384	10725	12066	13406	14747	16088	17428	20109
	1800	2.93		7313	8775	10238	11700	13163	14625	16088	17550	19013	21938
	1950	3.17		7922	9506	11091	12675	14259	15844	17428	19013	20597	23766
	2100	3.41		8531	10238	11944	13650	15356	17063	18769	20475	22181	25594
2275	1050	2.05	SEVEN	5119	6143	7166	8190	9214	10238	11261	12285	13309	15356
	1200	2.34		5850	7020	8190	9360	10530	11700	12870	14040	15210	17550
	1350	2.63		6581	7898	9214	10530	11846	13163	14479	15795	17111	19744
	1500	2.93		7313	8775	10238	11700	13163	14625	16088	17550	19013	21938
	1650	3.22		8044	9653	11261	12870	14479	16088	17696	19305	20914	24131
	1800	3.51		8775	10530	12285	14040	15795	17550	19305	21060	22815	26325
2275	1950	3.80	SEVEN	9506	11408	13309	15210	17111	19013	20914	22815	24716	28519
	2100	4.10		10238	12285	14333	16380	18428	20475	22523	24570	26618	30713
	2250	4.39		10969	13163	15356	17550	19744	21938	24131	26325	28519	32906
	1200	2.73		6825	8190	9555	10920	12285	13650	15015	16380	17745	20475
	1350	3.07		7678	9214	10749	12285	13821	15356	16892	18428	19963	23034
	1500	3.41		8531	10238	11944	13650	15356	17063	18769	20475	22181	25594
2275	1650	3.75	SEVEN	9384	11261	13138	15015	16892	18769	20646	22523	24399	28153
	1800	4.10		10238	12285	14333	16380	18428	20475	22523	24570	26618	30713
	1950	4.44		11091	13309	15527	17745	19963	22181	24399	26618	28836	33272
	2100	4.78		11944	14333	16721	19110	21499	23888	26276	28665	31054	35831
	2250	5.12		12797	15356	17916	20475	23034	25594	28153	30713	33272	38391
	2400	5.46		13650	16380	19110	21840	24570	27300	30030	32760	35490	40950

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths,

Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.89	X 0.94	X 1.00	X 1.04	X 1.10	X 1.15	X 1.20

For bend type attenuators 45% to be added to the pressure drop of the straight attenuator.

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-MA/20-150

Pressure Drop ΔP (Pa)				18	27	35	48	62	75	86	104	115	132
Face Velocity V_f (m/s)				3.0	4.0	4.5	5.4	6.0	6.5	7.0	7.5	8.0	9.0
Self generated noise level against face velocity V_f				NC 25			NC 30		NC 35		NC 40		NC 45
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)									
Width (mm)	Height (mm)												
350	150	0.05	ONE	158	210	236	284	315	341	368	394	420	473
	300	0.11		315	420	473	567	630	683	735	788	840	945
	450	0.16		473	630	709	851	945	1024	1103	1181	1260	1418
	600	0.21		630	840	945	1134	1260	1365	1470	1575	1680	1890
	750	0.26		788	1050	1181	1418	1575	1706	1838	1969	2100	2363
	900	0.32		945	1260	1418	1701	1890	2048	2205	2363	2520	2835
700	450	0.32	TWO	945	1260	1418	1701	1890	2048	2205	2363	2520	2835
	600	0.42		1260	1680	1890	2268	2520	2730	2940	3150	3360	3780
	750	0.53		1575	2100	2363	2835	3150	3413	3675	3938	4200	4725
	900	0.63		1890	2520	2835	3402	3780	4095	4410	4725	5040	5670
	1050	0.74		2205	2940	3308	3969	4410	4778	5145	5513	5880	6615
	1200	0.84		2520	3360	3780	4536	5040	5460	5880	6300	6720	7560
1050	600	0.63	THREE	1890	2520	2835	3402	3780	4095	4410	4725	5040	5670
	750	0.79		2363	3150	3544	4253	4725	5119	5513	5906	6300	7088
	900	0.95		2835	3780	4253	5103	5670	6143	6615	7088	7560	8505
	1050	1.10		3308	4410	4961	5954	6615	7166	7718	8269	8820	9923
	1200	1.26		3780	5040	5670	6804	7560	8190	8820	9450	10080	11340
	1350	1.42		4253	5670	6379	7655	8505	9214	9923	10631	11340	12758
1400	1500	1.58	FOUR	4725	6300	7088	8505	9450	10238	11025	11813	12600	14175
	750	0.05		3150	4200	4725	5670	6300	6825	7350	7875	8400	9450
	900	1.26		3780	5040	5670	6804	7560	8190	8820	9450	10080	11340
	1050	1.47		4410	5880	6615	7938	8820	9555	10290	11025	11760	13230
	1200	1.68		5040	6720	7560	9072	10080	10920	11760	12600	13440	15120
	1350	1.89		5670	7560	8505	10206	11340	12285	13230	14175	15120	17010
1750	1500	2.10	FIVE	6300	8400	9450	11340	12600	13650	14700	15750	16800	18900
	1650	2.31		6930	9240	10395	12474	13860	15015	16170	17325	18480	20790
	1800	2.52		7560	10080	11340	13608	15120	16380	17640	18900	20160	22680
	900	1.58		4725	6300	7088	8505	9450	10238	11025	11813	12600	14175
	1050	1.84		5513	7350	8269	9923	11025	11944	12863	13781	14700	16538
	1200	2.10		6300	8400	9450	11340	12600	13650	14700	15750	16800	18900
2100	1350	2.36	SIX	7088	9450	10631	12758	14175	15356	16538	17719	18900	21263
	1500	2.63		7875	10500	11813	14175	15750	17063	18375	19688	21000	23625
	1650	2.89		8663	11550	12994	15593	17325	18769	20213	21656	23100	25988
	1800	3.15		9450	12600	14175	17010	18900	20475	22050	23625	25200	28350
	1950	3.41		10238	13650	15356	18428	20475	22181	23888	25594	27300	30713
	2100	3.68		11025	14700	16538	19845	22050	23888	25725	27563	29400	33075
2450	1050	2.21	SEVEN	6615	8820	9923	11907	13230	14333	15435	16538	17640	19845
	1200	2.52		7560	10080	11340	13608	15120	16380	17640	18900	20160	22680
	1350	2.84		8505	11340	12758	15309	17010	18428	19845	21263	22680	25515
	1500	3.15		9450	12600	14175	17010	18900	20475	22050	23625	25200	28350
	1650	3.47		10395	13860	15593	18711	20790	22523	24255	25988	27720	31185
	1800	3.78		11340	15120	17010	20412	22680	24570	26460	28350	30240	34020
2450	1950	4.10	SEVEN	12285	16380	18428	22113	24570	26618	28665	30713	32760	36855
	2100	4.41		13230	17640	19845	23814	26460	28665	30870	33075	35280	39690
	2250	4.73		14175	18900	21263	25515	28350	30713	33075	35438	37800	42525
	1200	2.94		8820	11760	13230	15876	17640	19110	20580	22050	23520	26460
	1350	3.31		9923	13230	14884	17861	19845	21499	23153	24806	26460	29768
	1500	3.68		11025	14700	16538	19845	22050	23888	25725	27563	29400	33075
	1650	4.04		12128	16170	18191	21830	24255	26276	28298	30319	32340	36383
	1800	4.41		13230	17640	19845	23814	26460	28665	30870	33075	35280	39690
	1950	4.78		14333	19110	21499	25799	28665	31054	33443	35831	38220	42998
	2100	5.15		15435	20580	23153	27783	30870	33443	36015	38588	41160	46305
2450	2250	5.51	SEVEN	16538	22050	24806	29768	33075	35831	38588	41344	44100	49613
	2400	5.88		17640	23520	26460	31752	35280	38220	41160	44100	47040	52920

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths,

Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.90	X 0.95	X 1.00	X 1.05	X 1.11	X 1.16	X 1.19

For bend type attenuators 50% to be added to the pressure drop of the straight attenuator.

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-LM/20-175

Pressure Drop ΔP (Pa)			18	27	35	48	62	75	86	104	115	132	
Face Velocity V _f (m/s)			3.5	4.5	5.5	6.5	7.0	7.5	8.0	8.5	9.0	10.0	
Self generated noise level against face velocity V _f			NC 25			NC 30			NC 35		NC 40		NC 45
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)									
Width (mm)	Height (mm)												
375	150	0.06	ONE	197	253	309	366	394	450	478	506	534	563
	300	0.11		394	506	619	731	788	900	956	1013	1069	1125
	450	0.17		591	759	928	1097	1181	1350	1434	1519	1603	1688
	600	0.23		788	1013	1238	1463	1575	1800	1913	2025	2138	2250
	750	0.28		984	1266	1547	1828	1969	2250	2391	2531	2672	2813
	900	0.34		1181	1519	1856	2194	2363	2700	2869	3038	3206	3375
	1050	0.39		1378	1772	2166	2559	2756	3150	3347	3544	3741	3938
	1200	0.45		1575	2025	2475	2925	3150	3600	3825	4050	4275	4500
750	450	0.34	TWO	1181	1519	1856	2194	2363	2700	2869	3038	3206	3375
	600	0.45		1575	2025	2475	2925	3150	3600	3825	4050	4275	4500
	750	0.56		1969	2531	3094	3656	3938	4500	4781	5063	5344	5625
	900	0.68		2363	3038	3713	4388	4725	5400	5738	6075	6413	6750
	1050	0.79		2756	3544	4331	5119	5513	6300	6694	7088	7481	7875
	1200	0.90		3150	4050	4950	5850	6300	7200	7650	8100	8550	9000
	1350	1.01		3544	4556	5569	6581	7088	8100	8606	9113	9619	10125
	1500	1.03		3938	5063	6188	7313	7875	9000	9563	10125	10688	11250
1125	600	0.68	THREE	2363	3038	3713	4388	4725	5400	5738	6075	6413	6750
	750	0.84		2953	3797	4641	5484	5906	6750	7172	7594	8016	8438
	900	1.01		3544	4556	5569	6581	7088	8100	8606	9113	9619	10125
	1050	1.18		4134	5316	6497	7678	8269	9450	10041	10631	11222	11813
	1200	1.35		4725	6075	7425	8775	9450	10800	11475	12150	12825	13500
	1350	1.52		5316	6834	8353	9872	10631	12150	12909	13669	14428	15188
	1500	1.69		5906	7594	9281	10969	11813	13500	14344	15188	16031	16875
	1650	1.86		6497	8353	10209	12066	12994	14850	15778	16706	17634	18563
1500	1800	2.03	FOUR	7088	9113	11138	13163	14175	16200	17213	18225	19238	20250
	900	1.35		4725	6075	7425	8775	9450	10800	11475	12150	12825	13500
	1050	1.58		5513	7088	8663	10238	11025	12600	13388	14175	14963	15750
	1200	1.80		6300	8100	9900	11700	12600	14400	15300	16200	17100	18000
	1350	2.03		7088	9113	11138	13163	14175	16200	17213	18225	19238	20250
	1500	2.25		7875	10125	12375	14625	15750	18000	19125	20250	21375	22500
	1650	2.48		8663	11138	13613	16088	17325	19800	21038	22275	23513	24750
	1800	2.70		9450	12150	14850	17550	18900	21600	22950	24300	25650	27000
1875	1950	2.93	FIVE	10238	13163	16088	19013	20475	23400	24863	26325	27788	29250
	2100	3.15		11025	14175	17325	20475	22050	25200	26775	28350	29925	31500
	1050	1.97		6891	8859	10828	12797	13781	15750	16734	17719	18703	19688
	1200	2.25		7875	10125	12375	14625	15750	18000	19125	20250	21375	22500
	1350	2.53		8859	11391	13922	16453	17719	20250	21516	22781	24047	25313
	1500	2.81		9844	12656	15469	18281	19688	22500	23906	25313	26719	28125
	1650	3.09		10828	13922	17016	20109	21656	24750	26297	27844	29391	30938
	1800	3.38		11813	15188	18563	21938	23625	27000	28688	30375	32063	33750
2250	1950	3.66	SIX	12797	16453	20109	23766	25594	29250	31078	32906	34734	36563
	2100	3.94		13781	17719	21656	25594	27563	31500	33469	35438	37406	39375
	2250	4.22		14766	18984	23203	27422	29531	33750	35859	37969	40078	42188
	2400	4.50		15750	20250	24750	29250	31500	36000	38250	40500	42750	45000
	1200	2.70		9450	12150	14850	17550	18900	21600	22950	24300	25650	27000
	1350	3.04		10631	13669	16706	19744	21263	24300	25819	27338	28856	30375
	1500	3.38		11813	15188	18563	21938	23625	27000	28688	30375	32063	33750
	1650	3.71		12994	16706	20419	24131	25988	29700	31556	33413	35269	37125
2250	1800	4.05	SIX	14175	18225	22275	26325	28350	32400	34425	36450	38475	40500
	1950	4.39		15356	19744	24131	28519	30713	35100	37294	39488	41681	43875
	2100	4.73		16538	21263	25988	30713	33075	37800	40163	42525	44888	47250
	2250	5.06		17719	22781	27844	32906	35438	40500	43031	45563	48094	50625
	2400	5.40		18900	24300	29700	35100	37800	43200	45900	48600	51300	54000

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths,

Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.89	X 0.95	X 1.00	X 1.05	X 1.09	X 1.15	X 1.20

For bend type attenuators 58% to be added to the pressure drop of the straight attenuator.

Sound Attenuator

Aerodynamic Performance Data

Model: CSA-L/20-200

Pressure Drop ΔP (Pa)				18	27	35	48	62	75	86	104	115	132	
Face Velocity V_f (m/s)				4.0	5.0	6.0	7.3	8.0	9.0	9.5	10.5	11.0	11.5	
Self generated noise level against face velocity V_f				NC 25		NC 30		NC 35		NC 40		NC 45		NC 50
Attenuator Size		Face Area A (m ²)	No. of Splitters	Air Volume Q (L/s)										
Width (mm)	Height (mm)													
400	150	0.06	ONE	240	300	360	438	480	540	570	630	660	690	
	300	0.12		480	600	720	876	960	1080	1140	1260	1320	1380	
	450	0.18		720	900	1080	1314	1440	1620	1710	1890	1980	2070	
	600	0.24		960	1200	1440	1752	1920	2160	2280	2520	2640	2760	
	750	0.30		1200	1500	1800	2190	2400	2700	2850	3150	3300	3450	
	900	0.36		1440	1800	2160	2628	2880	3240	3420	3780	3960	4140	
	1050	0.42		1680	2100	2520	3066	3360	3780	3990	4410	4620	4830	
	1200	0.48		1920	2400	2880	3504	3840	4320	4560	5040	5280	5520	
800	450	0.36	TWO	1440	1800	2160	2628	2880	3240	3420	3780	3960	4140	
	600	0.48		1920	2400	2880	3504	3840	4320	4560	5040	5280	5520	
	750	0.60		2400	3000	3600	4380	4800	5400	5700	6300	6600	6900	
	900	0.72		2880	3600	4320	5256	5760	6480	6840	7560	7920	8280	
	1050	0.84		3360	4200	5040	6132	6720	7560	7980	8820	9240	9660	
	1200	0.96		3840	4800	5760	7008	7680	8640	9120	10080	10560	11040	
	1350	1.08		4320	5400	6480	7884	8640	9720	10260	11340	11880	12420	
	1500	1.20		4800	6000	7200	8760	9600	10800	11400	12600	13200	13800	
1200	600	0.72	THREE	2880	3600	4320	5256	5760	6480	6840	7560	7920	8280	
	750	0.90		3600	4500	5400	6570	7200	8100	8550	9450	9900	10350	
	900	1.08		4320	5400	6480	7884	8640	9720	10260	11340	11880	12420	
	1050	1.26		5040	6300	7560	9198	10080	11340	11970	13230	13860	14490	
	1200	1.44		5760	7200	8640	10512	11520	12960	13680	15120	15840	16560	
	1350	1.62		6480	8100	9720	11826	12960	14580	15390	17010	17820	18630	
	1500	1.80		7200	9000	10800	13140	14400	16200	17100	18900	19800	20700	
	1650	1.98		7920	9900	11880	14454	15840	17820	18810	20790	21780	22770	
1600	1800	2.16	FOUR	8640	10800	12960	15768	17280	19440	20520	22680	23760	24840	
	900	1.44		5760	7200	8640	10512	11520	12960	13680	15120	15840	16560	
	1050	1.68		6720	8400	10080	12264	13440	15120	15960	17640	18480	19320	
	1200	1.92		7680	9600	11520	14016	15360	17280	18240	20160	21120	22080	
	1350	2.16		8640	10800	12960	15768	17280	19440	20520	22680	23760	24840	
	1500	2.40		9600	12000	14400	17520	19200	21600	22800	25200	26400	27600	
	1650	2.64		10560	13200	15840	19272	21120	23760	25080	27720	29040	30360	
	1800	2.88		11520	14400	17280	21024	23040	25920	27360	30240	31680	33120	
2000	1950	3.12	FIVE	12480	15600	18720	22776	24960	28080	29640	32760	34320	35880	
	2100	3.36		13440	16800	20160	24528	26880	30240	31920	35280	36960	38640	
	1050	2.10		8400	10500	12600	15330	16800	18900	19950	22050	23100	24150	
	1200	2.40		9600	12000	14400	17520	19200	21600	22800	25200	26400	27600	
	1350	2.70		10800	13500	16200	19710	21600	24300	25650	28350	29700	31050	
	1500	3.00		12000	15000	18000	21900	24000	27000	28500	31500	33000	34500	
	1650	3.30		13200	16500	19800	24090	26400	29700	31350	34650	36300	37950	
	1800	3.60		14400	18000	21600	26280	28800	32400	34200	37800	39600	41400	
2400	1950	3.90	SIX	15600	19500	23400	28470	31200	35100	37050	40950	42900	44850	
	2100	4.20		16800	21000	25200	30660	33600	37800	39900	44100	46200	48300	
	2250	4.50		18000	22500	27000	32850	36000	40500	42750	47250	49500	51750	
	2400	4.80		19200	24000	28800	35040	38400	43200	45600	50400	52800	55200	
	1200	2.88		11520	14400	17280	21024	23040	25920	27360	30240	31680	33120	
	1350	3.24		12960	16200	19440	23652	25920	29160	30780	34020	35640	37260	
	1500	3.60		14400	18000	21600	26280	28800	32400	34200	37800	39600	41400	
	1650	3.96		15840	19800	23760	28908	31680	35640	37620	41580	43560	45540	
2400	1800	4.32	SIX	17280	21600	25920	31536	34560	38880	41040	45360	47520	49680	
	1950	4.68		18720	23400	28080	34164	37440	42120	44460	49140	51480	53820	
	2100	5.04		20160	25200	30240	36792	40320	45360	47880	52920	55440	57960	
	2250	5.40		21600	27000	32400	39420	43200	48600	51300	56700	59400	62100	
	2400	5.76		23040	28800	34560	42048	46080	51840	54720	60480	63360	66240	

Above pressure loss data is related to a straight attenuator of 1200mm long. The following correction factors to be applied for other lengths,
Pressure Drop correction table:

Attenuator Length (mm)	600	900	1200	1500	1800	2100	2400
Pressure Drop correction factor (ΔP)	X 0.90	X 0.96	X 1.00	X 1.04	X 1.11	X 1.15	X 1.19

For bend type attenuators 65% to be added to the pressure drop of the straight attenuator.

Sound Attenuators

Dynamic Insertion Loss

Attenuator Model	Length in mm	Octave Band Center Frequency fm in Hz / Dynamic Insertion Loss De in dB							
		63	125	250	500	1K	2K	4K	8K
CSA-H/20-75	600	7	11	16	30	40	34	28	20
	900	9	13	24	38	48	46	35	26
	1200	10	18	30	45	50	50	45	38
	1500	12	20	36	50	50	50	48	43
	1800	13	24	42	50	50	50	50	46
	2100	15	27	46	50	50	50	50	49
	2400	18	31	50	50	50	50	50	50
CSA-HM/20-100	600	6	9	14	21	34	26	23	16
	900	8	11	20	30	44	38	30	21
	1200	9	14	24	39	49	48	36	28
	1500	11	17	29	45	50	50	42	32
	1800	12	20	34	49	50	50	46	38
	2100	14	22	39	50	50	50	48	40
	2400	15	24	44	50	50	50	50	46
CSA-M/20-125	600	5	8	13	19	27	22	18	14
	900	7	10	18	27	38	30	23	18
	1200	8	12	20	34	45	43	31	23
	1500	9	14	25	42	49	45	34	27
	1800	10	17	27	46	50	49	38	32
	2100	12	20	33	49	50	50	42	35
	2400	13	22	36	50	50	50	46	38
CSA-MA/20-150	600	5	7	12	18	24	20	15	12
	900	6	9	16	24	34	26	18	16
	1200	7	10	18	32	43	41	29	20
	1500	8	13	22	39	48	43	31	24
	1800	9	15	24	44	50	46	34	28
	2100	10	18	30	49	50	49	39	32
	2400	12	20	33	50	50	50	43	35
CSA-LM/20-175	600	3	5	9	14	20	17	13	11
	900	4	6	11	20	29	23	16	13
	1200	5	7	14	25	41	40	27	18
	1500	6	9	17	31	45	42	29	21
	1800	7	11	21	37	49	45	32	24
	2100	8	14	26	42	50	49	36	28
	2400	10	16	29	48	50	50	39	30
CSA-L/20-200	600	2	3	8	12	18	15	11	10
	900	3	4	9	18	25	20	14	12
	1200	2	4	10	22	39	39	26	17
	1500	4	7	14	27	42	40	28	19
	1800	6	9	18	33	47	43	31	22
	2100	7	12	24	38	50	47	35	26
	2400	9	15	27	46	50	49	37	28

Insertion Loss correction table for Melinex Acoustic infill:

Multiplication Factor	X 0.98	X 0.82	X 0.83	X 0.94	X 0.96	X 0.72	X 0.65	X 0.82
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Sound Attenuators

Weight Chart

Model: CSA-H/20-75

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size		Attenuator Weight in Kg (+/-10%)						
Width (mm)	Height (mm)	No. of Splitters						
275	300	One	18	22	27	31	36	40
	600		23	30	37	44	51	58
	900		29	38	47	57	66	75
	1200		34	46	57	69	81	93
	1500		39	53	68	82	96	110
550	300	Two	45	61	78	95	111	128
	600		26	34	41	49	57	65
	900		34	46	58	70	82	94
	1200		42	59	75	91	107	123
	1500		51	71	91	112	132	152
825	300	Three	59	84	108	132	157	181
	600		68	96	124	153	181	210
	900		33	45	56	68	79	91
	1200		45	62	80	97	114	131
	1500		57	80	103	126	149	172
1100	300	Four	69	97	126	155	183	212
	600		80	115	149	184	218	253
	900		92	132	173	213	253	293
	1200		41	56	71	87	102	117
	1500		57	79	101	124	146	169
1375	300	Five	72	102	131	161	191	221
	600		87	124	162	199	236	273
	900		103	147	192	236	280	325
	1200		118	170	222	273	325	377
	1500		49	68	87	105	124	143
1650	300	Six	69	96	124	152	179	207
	600		88	124	161	198	234	271
	900		107	153	198	244	289	335
	1200		126	181	235	290	344	399
	1500		145	209	272	336	400	463
1925	300	Seven	58	80	102	125	147	169
	600		81	114	147	180	213	245
	900		104	148	191	235	278	322
	1200		128	182	236	290	344	398
	1500		151	216	280	345	410	474
2200	300	Eight	174	249	325	400	475	551
	600		66	92	118	144	170	196
	900		94	132	170	208	246	284
	1200		122	172	222	273	323	373
	1500		149	212	274	337	399	462
2475	300	Nine	177	252	326	401	476	551
	600		204	291	378	465	553	640
	900		75	105	134	164	193	222
	1200		107	151	194	237	281	324
	1500		140	197	254	311	368	425
	300		172	243	314	385	456	527
	600		204	289	374	459	544	628
	900		236	335	434	532	631	730
	1200		84	117	150	183	217	250
	1500		121	170	218	267	315	364
	300		158	222	286	350	414	478
	600		195	275	354	434	513	593
	900		232	327	422	517	612	707
	1200		269	380	490	601	711	822
	1500							
	1800							

Model: CSA-HM/20-100

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size		Attenuator Weight in Kg (+/-10%)						
Width (mm)	Height (mm)	No. of Splitters						
300	300	One	17	21	26	30	34	38
	600		22	29	35	42	49	55
	900		27	36	45	54	63	72
	1200		32	44	55	66	78	89
	1500		41	51	65	79	92	106
600	300	Two	42	58	75	91	107	123
	600		24	32	40	48	56	64
	900		33	44	56	68	80	92
	1200		41	57	72	88	104	120
	1500		49	69	89	108	128	148
900	300	Three	57	81	105	129	152	176
	600		65	93	121	149	176	204
	900		32	44	55	66	78	89
	1200		44	61	78	95	112	129
	1500		55	78	100	123	146	168
1200	300	Four	67	95	123	151	180	208
	600		78	112	146	180	214	247
	900		90	129	169	208	248	287
	1200		40	55	70	85	100	115
	1500		55	77	100	122	144	166
1500	300	Five	71	100	129	159	188	217
	600		86	122	159	195	232	269
	900		101	145	189	232	276	320
	1200		116	167	218	269	320	371
	1500		48	67	85	104	122	141
1800	300	Six	67	95	122	149	177	204
	600		87	123	159	195	231	267
	900		106	151	196	241	286	331
	1200		125	179	232	286	340	394
	1500		158	207	269	332	394	457
2100	300	Seven	62	79	101	123	145	167
	600		80	113	145	177	210	242
	900		103	146	189	232	275	318
	1200		127	180	233	287	340	393
	1500		150	214	277	341	405	469
2400	300	Eight	173	247	322	396	470	544
	600		65	91	117	142	168	193
	900		93	131	168	206	244	281
	1200		121	171	220	270	320	369
	1500		149	204	272	334	396	457
	300		179	250	324	398	472	545
	600		204	290	376	462	548	633
	900		74	104	133	162	191	220
	1200		107	150	192	235	278	321
	1500		139	196	252	309	365	422
	300		172	242	312	382	452	522
	600		204	288	372	455	539	623
	900		236	334	431	529	626	724
	1200							
	1500							

*Total Weight of units larger than given can be obtained by summation of above weights.

*If the attenuator dimension exceeds Width (W) = 2100mm, Height (H) = 1800mm and Length (L) = 2100mm would be supplied in multiple sections or banks.

Sound Attenuators

Weight Chart

Model: CSA-M/20-125

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size Width (mm)	Height (mm)	No. of Splitters	Attenuator Weight in Kg (+/-10%)					
325	300	One	17	21	25	30	34	39
	600		22	29	35	42	49	55
	900		27	36	45	54	63	72
	1200		32	43	55	66	78	89
	1500		37	51	65	78	92	106
	1800		42	58	74	91	107	123
650	300	Two	24	32	40	48	56	64
	600		33	45	57	69	81	93
	900		41	57	73	89	105	121
	1200		49	69	90	110	130	150
	1500		58	82	106	130	154	178
	1800		66	94	122	150	179	207
975	300	Three	32	44	56	67	79	91
	600		44	62	79	96	114	131
	900		56	79	102	125	148	172
	1200		68	97	126	154	183	212
	1500		80	114	149	183	218	252
	1800		92	132	172	212	252	293
1300	300	Four	41	56	71	87	102	117
	600		56	79	102	124	147	170
	900		72	102	132	162	192	222
	1200		88	125	163	200	237	275
	1500		103	148	193	238	283	327
	1800		119	171	223	276	328	380
1625	300	Five	49	68	87	106	125	144
	600		69	97	125	153	181	209
	900		89	126	163	200	237	274
	1200		109	155	201	247	293	339
	1500		128	183	239	294	349	404
	1800		148	212	276	340	405	469
1950	300	Six	58	81	103	126	148	171
	600		82	116	149	182	215	249
	900		106	150	194	238	282	326
	1200		131	185	240	295	349	404
	1500		155	220	286	351	416	482
	1800		179	255	331	407	483	559
2275	300	Seven	67	93	120	146	172	198
	600		96	135	173	212	250	289
	900		125	176	227	278	329	380
	1200		154	217	280	344	407	470
	1500		183	258	334	410	485	561
	1800		211	299	387	476	564	652

Model: CSA-MA/20-150

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size Width (mm)	Height (mm)	No. of Splitters	Attenuator Weight in Kg (+/-10%)					
350	300	One	16	21	25	30	34	38
	600		21	28	35	42	48	55
	900		26	36	45	54	63	72
	1200		31	43	54	66	77	89
	1500		36	50	64	78	92	106
	1800		41	58	74	90	106	123
700	300	Two	24	32	41	49	57	65
	600		33	45	57	69	81	94
	900		41	57	74	90	106	122
	1200		49	70	90	110	131	151
	1500		58	82	107	131	155	180
	1800		66	95	123	152	180	209
1050	300	Three	32	44	56	68	80	92
	600		45	62	80	97	115	133
	900		57	80	103	127	150	173
	1200		69	98	127	156	185	214
	1500		81	115	150	185	220	255
	1800		93	133	174	215	255	296
1400	300	Four	41	57	72	88	103	119
	600		57	80	103	126	149	172
	900		73	103	134	164	195	225
	1200		89	127	165	203	241	279
	1500		105	150	196	241	287	332
	1800		121	174	227	280	332	385
1750	300	Five	50	69	88	108	127	146
	600		70	99	127	156	184	212
	900		90	128	166	203	241	279
	1200		111	158	204	251	298	345
	1500		131	187	243	299	355	411
	1800		151	217	282	347	412	477
2100	300	Six	59	82	105	128	151	174
	600		84	118	152	185	219	253
	900		109	154	198	243	288	332
	1200		134	189	245	300	356	412
	1500		159	225	292	358	424	491
	1800		184	261	338	415	493	570
2450	300	Seven	68	95	122	149	175	202
	600		98	137	177	216	255	295
	900		128	180	232	284	335	387
	1200		158	222	287	351	415	480
	1500		188	265	342	419	495	572
	1800		218	307	397	486	576	665

*Total Weight of units larger than given can be obtained by summation of above weights.

*If the attenuator dimension exceeds Width (W) = 2100mm, Height (H) = 1800mm and Length (L) = 2100mm would be supplied in multiple sections or banks.

Sound Attenuators

Weight Chart

Model: CSA-LM/20-175

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size Width (mm)	Height (mm)	No. of Splitters	Attenuator Weight in Kg (+/-10%)					
375	300	One	16	21	25	30	34	38
	600		21	28	35	42	48	55
	900		26	35	45	54	63	72
	1200		31	43	54	66	77	89
	1500		36	50	64	78	92	106
	1800		41	58	74	90	106	123
750	300	Two	24	33	41	49	57	66
	600		33	45	58	70	82	95
	900		41	58	74	91	107	124
	1200		50	70	91	112	132	153
	1500		58	83	108	132	157	182
	1800		67	96	124	153	182	211
1125	300	Three	33	45	57	69	81	93
	600		45	63	81	99	117	135
	900		57	81	105	129	153	176
	1200		70	99	129	159	188	218
	1500		82	117	153	189	224	260
	1800		94	136	177	218	260	301
1500	300	Four	42	57	73	89	105	121
	600		58	81	105	128	152	175
	900		74	106	137	168	199	230
	1200		91	130	168	207	246	284
	1500		107	154	200	246	292	339
	1800		124	178	231	285	339	393
1875	300	Five	51	70	90	110	130	149
	600		72	101	130	159	188	217
	900		93	131	169	208	246	285
	1200		114	161	209	257	304	353
	1500		135	192	249	306	363	420
	1800		155	222	288	355	421	487
2250	300	Six	60	84	107	131	154	178
	600		86	121	155	190	224	259
	900		112	157	203	249	294	340
	1200		138	194	251	308	364	421
	1500		163	231	299	367	435	502
	1800		189	268	347	426	505	583

Model: CSA-L/20-200

Attenuator Length (mm)		600	900	1200	1500	1800	2100	2400
Attenuator Size Width (mm)	Height (mm)	No. of Splitters	Attenuator Weight in Kg (+/-10%)					
400	300	One	16	21	25	30	34	39
	600		21	28	35	42	49	56
	900		26	36	45	54	64	73
	1200		31	43	55	67	78	90
	1500		36	51	65	79	93	107
	1800		41	58	75	91	108	124
800	300	Two	25	33	41	50	58	67
	600		33	46	58	71	84	96
	900		42	59	76	92	109	126
	1200		50	71	93	114	135	156
	1500		59	84	110	135	160	185
	1800		68	97	127	156	185	215
1200	300	Three	33	46	58	70	83	95
	600		46	64	83	101	119	138
	900		59	83	107	132	156	180
	1200		71	102	132	162	193	223
	1500		84	120	156	193	229	266
	1800		96	139	181	223	266	308
1600	300	Four	42	59	75	91	108	124
	600		59	84	108	132	156	180
	900		76	108	140	172	204	236
	1200		93	133	173	212	252	292
	1500		110	158	205	253	300	348
	1800		127	183	238	293	348	404
2000	300	Five	52	72	92	113	133	153
	600		74	103	133	163	193	223
	900		95	135	174	213	253	292
	1200		117	166	215	264	313	362
	1500		139	197	256	314	373	431
	1800		161	229	297	365	433	501
2400	300	Six	62	86	110	134	158	183
	600		89	124	160	195	231	266
	900		116	162	209	256	303	350
	1200		142	201	259	317	375	434
	1500		169	239	308	378	448	517
	1800		196	277	358	439	520	601

*Total Weight of units larger than given can be obtained by summation of above weights.

*If the attenuator dimension exceeds Width (W) = 2100mm, Height (H) = 1800mm and Length (L) = 2100mm would be supplied in multiple sections or banks.

Sound Attenuators

Self Generated Noise

Model: CSA-H/20-75

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	2.5	54	43	37	30	27	24	20	20
	3.8	58	46	41	33	30	27	24	22
	5.0	62	55	48	42	38	35	34	32
900	2.5	54	42	36	31	26	23	22	20
	3.8	58	47	42	33	30	27	24	22
	5.0	63	55	49	43	39	35	34	32
1200	2.5	54	42	36	31	28	23	23	21
	3.8	59	47	42	34	31	27	24	22
	5.0	64	57	50	45	41	39	38	37
1500	2.5	55	43	37	32	28	24	22	21
	3.8	59	47	43	34	32	29	24	24
	5.0	64	58	51	45	40	38	36	35
1800	2.5	55	43	37	31	29	24	22	20
	3.8	59	47	44	35	32	29	25	24
	5.0	64	59	52	46	41	39	36	35
2100	2.5	55	42	37	32	28	25	22	21
	3.8	58	48	44	35	33	28	24	24
	5.0	65	57	50	45	41	39	38	36
2400	2.5	54	41	37	31	29	25	22	21
	3.8	58	47	43	35	32	28	25	24
	5.0	65	55	51	45	40	39	37	35

Model: CSA-HM/20-100

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	3.0	53	42	36	30	29	25	23	21
	4.5	59	47	41	34	31	28	25	23
	6.0	60	51	44	39	34	33	33	30
900	3.0	54	42	37	31	28	25	24	22
	4.5	60	48	42	35	31	29	25	23
	6.0	60	51	43	38	35	33	32	31
1200	3.0	54	43	37	31	28	24	22	21
	4.5	59	48	40	34	30	28	25	22
	6.0	61	52	45	40	35	34	33	31
1500	3.0	54	44	37	30	29	25	23	22
	4.5	60	48	40	34	32	28	25	23
	6.0	61	52	44	41	35	34	34	32
1800	3.0	54	44	38	31	30	26	23	21
	4.5	60	48	40	34	33	28	26	24
	6.0	60	52	45	40	35	34	33	32
2100	3.0	55	45	38	32	30	26	24	22
	4.5	60	49	41	35	33	29	26	25
	6.0	60	52	45	40	35	34	33	31
2400	3.0	55	46	38	33	31	27	25	22
	4.5	61	49	40	35	34	28	27	24
	6.0	60	52	45	40	36	34	33	32

*Above self generated noise is based on both noise and airflow are travelling in the same direction.

Sound Attenuators

Self Generated Noise

Model: CSA-M/20-125

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	3.5	53	43	36	31	28	24	22	20
	5.0	58	47	40	34	30	26	24	22
	7.0	64	57	50	44	41	38	38	36
900	3.5	53	44	36	31	28	25	22	21
	5.0	58	48	40	34	30	27	25	22
	7.0	64	57	50	45	42	39	38	35
1200	3.5	53	44	36	31	27	24	22	21
	5.0	59	48	41	34	30	27	25	23
	7.0	64	57	50	45	41	39	38	37
1500	3.5	53	43	36	30	27	24	23	22
	5.0	58	48	40	34	31	28	25	23
	7.0	64	56	50	45	40	38	37	36
1800	3.5	54	44	37	31	27	24	22	21
	5.0	58	49	41	34	31	29	25	23
	7.0	64	57	51	45	41	39	37	36
2100	3.5	53	44	36	32	27	24	22	21
	5.0	58	48	40	34	30	27	25	22
	7.0	64	56	50	45	41	38	37	35
2400	3.5	53	45	35	32	28	25	23	22
	5.0	58	49	41	35	30	27	25	23
	7.0	64	56	50	44	41	39	37	34

Model: CSA-MA/20-150

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	4.0	54	43	35	31	28	24	22	20
	5.5	58	48	41	34	30	28	25	23
	7.5	60	52	46	41	36	34	32	31
900	4.0	54	44	36	30	28	24	23	20
	5.5	58	47	40	35	30	28	26	24
	7.5	61	52	47	42	37	34	32	30
1200	4.0	55	44	36	31	28	24	22	21
	5.5	59	48	41	34	30	28	25	23
	7.5	60	52	46	40	36	34	32	32
1500	4.0	53	44	37	30	26	24	22	20
	5.5	58	47	40	33	31	27	26	23
	7.5	61	52	45	40	36	34	33	31
1800	4.0	53	44	37	30	26	25	22	21
	5.5	59	47	40	32	32	28	26	24
	7.5	60	52	45	40	36	34	34	32
2100	4.0	54	45	36	31	27	24	22	21
	5.5	59	47	40	33	31	28	26	23
	7.5	61	53	46	41	36	34	33	33
2400	4.0	55	46	36	31	28	24	23	22
	5.5	60	47	41	32	31	28	27	24
	7.5	62	54	46	41	37	34	33	31

*Above self generated noise is based on both noise and airflow are travelling in the same direction.

Sound Attenuators

Self Generated Noise

Model: CSA-LM/20-175

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	4.5	54	45	37	31	28	25	22	20
	6.0	59	48	40	34	30	28	25	22
	8.0	62	52	45	41	37	33	32	31
900	4.5	54	45	38	32	29	25	22	21
	6.0	60	49	41	35	31	29	25	23
	8.0	62	53	45	41	38	34	32	30
1200	4.5	54	45	37	32	28	25	23	21
	6.0	59	49	40	34	31	28	25	23
	8.0	62	52	45	41	37	34	33	32
1500	4.5	54	45	38	32	29	25	23	22
	6.0	59	50	41	34	31	28	25	24
	8.0	65	52	46	41	37	35	34	32
1800	4.5	55	45	38	31	30	25	24	22
	6.0	60	51	42	34	31	29	26	24
	8.0	65	52	45	42	38	34	34	32
2100	4.5	54	44	37	33	30	25	24	21
	6.0	61	50	42	34	30	28	27	23
	8.0	66	53	47	42	39	34	33	31
2400	4.5	55	45	38	34	31	26	25	22
	6.0	60	52	43	35	31	29	28	25
	8.0	66	54	47	41	39	33	32	31

Model: CSA-L/20-200

Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1K	2K	4K	8K
600	5.0	54	43	36	30	27	25	22	21
	7.5	58	47	39	32	30	28	26	22
	10.0	64	58	50	53	52	52	46	37
900	5.0	54	44	37	31	27	24	22	21
	7.5	50	47	41	34	30	28	26	22
	10.0	64	58	51	46	40	39	38	36
1200	5.0	55	47	42	44	44	39	31	29
	7.5	59	53	49	54	53	52	46	37
	10.0	64	60	56	56	58	59	56	49
1500	5.0	55	46	37	39	28	24	23	21
	7.5	59	48	41	35	30	28	25	23
	10.0	67	60	54	48	45	43	41	40
1800	5.0	55	45	38	32	29	25	24	22
	7.5	59	49	42	35	31	29	25	23
	10.0	68	61	55	49	45	44	42	41
2100	5.0	54	46	38	33	28	26	23	22
	7.5	58	50	42	36	32	30	25	23
	10.0	65	61	54	50	44	43	41	40
2400	5.0	54	44	37	31	28	25	22	21
	7.5	59	48	41	33	30	27	25	23
	10.0	65	57	50	45	41	29	38	36

*Above self generated noise is based on both noise and airflow are travelling in the same direction.

Sound Attenuators

Acoustic Design

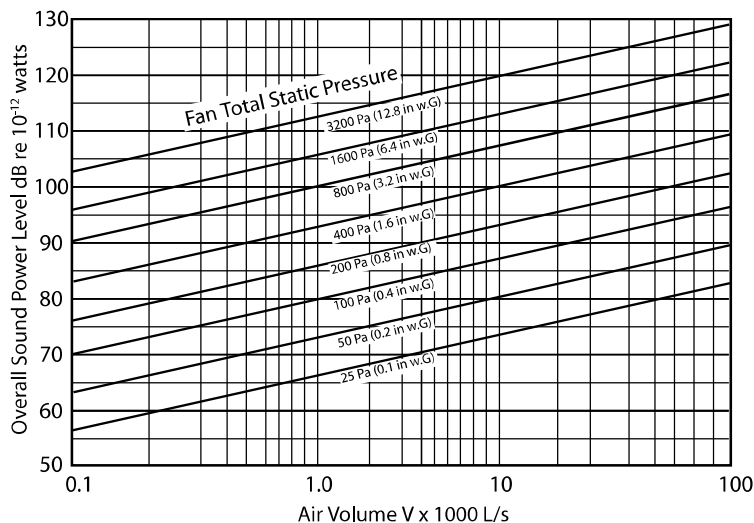
Acoustic Design of Sound Attenuator (Room Side Calculation)

A system analysis should be carried out on all HVAC systems having ducted connections to the area under consideration.

In-duct SWL of the fan

Using the fan manufacturer's data sheet (catalogue) obtain the in-duct sound power level at the Octave band centre frequencies. Or the approximate Octave band sound power level can be found from the below Airflow Vs Sound power level Table-1a at a specified fan static pressure. In both the cases the approximate duty of the fan needs to be known. These values are inserted against the source sound power level in Line-A.

Fan Induct SWL Table (Table-1a):



Spectrum Correction (Table-1b):

Octave Band Center Frequency in Hz	63	125	250	500	1K	2K	4K	8K
Forward curved centrifugal fan	-2	-7	-12	-17	-22	-27	-32	-37
Backward curved centrifugal fan	-7	-8	-7	-12	-17	-22	-27	-32
Axial fan	-5	-5	-6	-7	-8	-11	-14	-17

*Some manufactures present noise data as sound pressure level which needs to be converted by applying the relevant correction factor.

Sound Attenuators

Acoustic Design

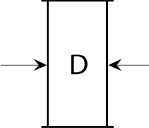
Sound Power Level (SWL) Leaving the System

Investigate the duct system between the fan and the critical noise outlet. The critical noise outlet in the duct system is the outlet closest to the sound source or fan. Using the following tables we can assess the total duct attenuation.

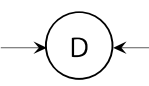
Attenuation of Rectangular Ducts

The straight unlined duct offers a degree of attenuation. This is frequency depends and varies with the duct dimension and duct length, which should be taken into account. Hence determining the length and sizes of all sections of duct is a must. Attenuation per meter of straight unlined rectangular duct is given in Table-2a and circular (round) duct is given in Table-2b. This should be inserted in Line-B.

Attenuation of straight unlined rectangular sheet metal ducts - dB/m (Table-2a):

	Small duct dimension D in mm	Octave band centre frequency fm in Hz							
		63	125	250	500	1k	2k	4k	8k
	0 - 200	0.6	0.6	0.45	0.3	0.3	0.3	0.3	0.3
	201 - 400	0.6	0.6	0.45	0.3	0.2	0.2	0.2	0.2
	401 - 800	0.6	0.6	0.3	0.15	0.15	0.15	0.15	0.15
	801 - 1600	0.3	0.15	0.15	0.1	0.06	0.06	0.06	0.06

Attenuation of straight unlined circular/round sheet metal ducts - dB/m (Table-2b):

	Duct dimension D in mm	Octave band centre frequency fm in Hz							
		63	125	250	500	1k	2k	4k	8k
	0 - 180	0.03	0.03	0.05	0.05	0.1	0.1	0.1	0.1
	181 - 380	0.03	0.03	0.03	0.05	0.07	0.07	0.07	0.07
	381 - 760	0.02	0.02	0.02	0.03	0.05	0.05	0.05	0.05
	761 - 1520	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02

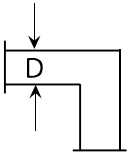
Sound Attenuators

Acoustic Design

Attenuation of Mitered Bends

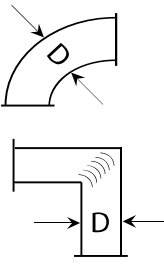
Bends with short chord turning vanes or without turning vanes provide considerable attenuation. Table-3a provides the dB level of the bend without turning vanes. This should be inserted in Line-B.

Attenuation of mitered bends without turning vanes or with short chord turning vanes (Table-3a):

	Duct Dimensions D in mm	Octave band centre frequency fm in Hz							
		63	125	250	500	1K	2K	4K	8K
	0 - 200	0	0	0	0	6	8	4	3
	201 - 400	0	0	0	6	8	4	3	3
	401 - 800	0	0	6	8	4	3	3	3
	801 - 2000	0	6	8	4	3	3	3	3

Bends with long chord turning vanes and radiussed bends provides a degree of attenuation which is given in the below Table-3b. This should be inserted in the Line-B.

Attenuation of radiussed circular / rectangular bends or mitered bends with long chord turning vanes (Table-3b):

	Dimension D in mm	Octave band centre frequency fm in Hz							
		63	125	250	500	1K	2K	4K	8K
	0 - 250	0	0	0	0	1	2	3	3
	251 - 500	0	0	0	1	2	3	3	3
	501 - 1000	0	0	1	2	3	3	3	3
	1001 - 2000	0	1	2	3	3	3	3	3

Sound Attenuators

Acoustic Design

Outlet Reflection

The sound power reaching the critical noise outlet is reflected back along the duct. The degree of attenuation due to this phenomenon is depend on frequency and the total area of the outlet or size of the outlet. Attenuation from Table-4 should be inserted in Line-C.

Outlet reflection in dB (Table - 4):

Octave band centre frequency fm in Hz	1K	3	2	1				0										
	500	6	5	4	3	2		1			0							
	250	10	9	8	7	6	5	4	3	2	1	0						
	125	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1		
	63	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4
Outlet Area in cm²	100 </																	

The total duct attenuation is obtained from adding Line-B and C. The sound power level leaving the critical outlet is obtained from Line-E=Line A-Line (B+C).

Room Effect

Sound pressure waves will reach the listener along two paths in a room, they are;

1. Directly, reducing as the (distance)² from the noise source, known as the Direct Sound Pressure level.
2. By multiple reflections off the room surfaces and room contents (furniture), which will depend upon the size of the room and the reverberation time, known as the Reverberant Sound Pressure Level.

Direct Sound Pressure Level

Calculate the percentage of the total sound leaving the critical noise outlet. This is approximately equal to the percentage of the total fan air volume which passes through the selected critical noise outlet for the analysis. Table-5 represents the values to be inserted in line-F.

Percentage Leaving Outlet

Percentage leaving the outlet Vs Total sound pressure level in dB (Table - 5):

dB	-20	-19	-18	-17	-16	-15	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
	1			2		3	4	5		10		20						50			100
Percentage leaving the outlet (%)																					

Sound Attenuators

Acoustic Design

Distance Factor

Estimate the distance between the nearest listening position and the critical noise source. From Table-6, the distance factor to be inserted in Line-G. If the distance is unknown, 1.5m can be considered as standard.

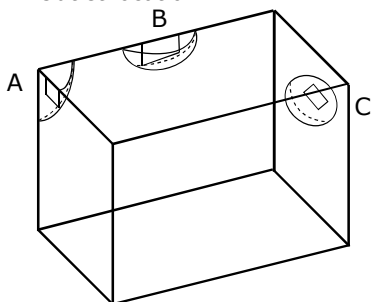
Distance Factor in dB (Table - 6):

dB	-10	-11	-12	-13	-14	-15	-16	-17	-18	-19	-20	-21	-22	-23	-24	-25	-26	-27	-28	-29	-30
		1			1.5		2		3		4		5		6		7		8		9
Distance in metres (m)																					

Directivity Factor

Select the location of the outlet (Type-A or Type-B or Type-C) to find the directivity factor in dB. Type-A indicates the outlet is at the junction of three wall or room surfaces. Type-B indicates the outlet is at the junction of two wall or room surfaces. Type-C indicates the outlet is at the centre of wall or room surfaces or at the ceiling of the room. Form the Tables 7a, 7b & 7c, the relevant values to be inserted in line-H.

Outlet location:



Type - A: Junction of three room surfaces (Table - 7a):

Type - A	Octave band center frequency fm in Hz							
	63	125	250	500	1K	2K	4K	8K
	+9	+9	+9	+9	+9	+9	+9	+9

Type - B: Junction of two room surfaces (Table - 7b):

Outlet Area in cm ²	Octave band center frequency in Hz							
	63	125	250	500	1K	2K	4K	8K
10000	+8	+8	+9	+9	+9	+9	+9	+9
1000	+7	+7	+8	+8	+8	+8	+8	+8
100	+6	+6	+7	+7	+7	+7	+7	+7
10	+5	+5	+6	+6	+6	+6	+6	+6

Type - C: Center of wall or Ceiling surface (Table - 7c):

Outlet Area in cm ²	Octave band center frequency in Hz							
	63	125	250	500	1K	2K	4K	8K
100000	+8	+8	+9	+9	+9	+9	+9	+9
10000	+7	+7	+8	+8	+8	+8	+8	+8
1000	+6	+6	+7	+7	+7	+7	+7	+7
100	+5	+5	+6	+6	+6	+6	+6	+6
10	+4	+4	+5	+5	+5	+5	+5	+5

The direct sound pressure level in the room is equal to the sum of the sound power level leaving the critical outlet in Line-E and the total of direct sound factors from Line F, G and H. This is inserted in Line-I

Sound Attenuators

Acoustic Design

Combining Two Sound Levels

To arrive the combined sound pressure level, it is necessary to logarithmically sum the reverberant sound pressure level and the direct sound pressure level. This can be simplified by using Table-11. This value can be entered in Line-P.

Combining two sound levels (Table 11):

Addition of Sound Levels, dB	
Difference in Sound Level	Value to be added to Larger Sound Level
0, 1	+3
2, 3	+2
4, 5, 6, 7, 8, 9	+1
10 & Above	+0

Required Insertion Loss

Required noise criterion can be selected from the Table-12 and the corresponding sound pressure level values can be taken from Table-13a or 13b or 13c and it should be inserted in Line-Q.

If the combined sound pressure level exceeds the criterion in any octave band, then the difference is the Insertion loss required from the attenuator (Line-R).

Safety margin of 3dB can be added for the possible addition of noise from other sources. The selected insertion loss figures are placed in Line-S as a final check.

Nomenclature

W - Width in mm

H - Height in mm

L - Length in mm

D - Splitter Thickness in mm.

S-Airway width in mm.

Q-Air Volume in L/s.

V_f -Face Velocity in m/s ($V_f=Q/(W \times H)$).

ΔP -Pressure Loss in Pa.

f_m -Octave Band Centre Frequency in Hz.

D_e -Dynamic Insertion Loss in dB.

NC-Noise Criterion.

Sound Attenuators

Acoustic Design

Recommended NC Level for Various Activities or Occupied Spaces (Table 12):

Room Type	Recommended NC Level
Private Residences	
Bed rooms	25 - 30
Apartments	30 - 40
Family Rooms & Living Rooms	30 - 40
Hotels and Motels	
Individual Rooms or Suites	25 - 35
Meeting / Ball Rooms / Banquet Rooms	30 - 35
Corridors and Lobbies	35 - 45
Service, Support Areas	35 - 45
Kitchen, Laundry	35 - 45
Restaurants, Shops and Stores	
Restaurants, Department Stores	35 - 40
Clubs, Public Houses, Cafeteria, Canteens and Retail Stores	40 - 45
Office Buildings	
Executive, Private Offices	25 - 35
Conference Rooms, Reception	30 - 35
Teleconference Rooms	20 - 25
Drawing Office, Computer Rooms	35 - 45
Corridors and Lobbies	40 - 45
Public Buildings	
Assembly Hall	25 - 35
Court Rooms	25 - 30
Public Library, Bank and Museum	30 - 35
Wash rooms, Toilets	35 - 45
Swimming Pool, Sports Area	40 - 50
Garage, Car Park	55
Laboratories	
Testing and Research	45 - 55
Group Teaching	35 - 45
General Assembly	25 - 35

(Cont....)

Sound Attenuators

Acoustic Design

Recommended NC Level (Table 12):

(Cont....)

Room Type	Recommended NC Level
Industrial	
Warehouse, Garage	45 - 50
Light Engineering workshop	45 - 55
Heavy Engineering workshop	50 - 65
Performing Arts Spaces (Studios and Auditoriums)	
Sound Broadcasting, TV, Recording Studio	20
Drama Theatres and Music Studios	20 - 25
Music Practice Rooms, Audience Studio	25 - 30
Concert Halls and Opera Houses	20 - 25
Lecture Theatre, Cinema Theatre	25 - 30
Stadium	
Indoor Stadiums	40 - 50
Large Seating Stadium	45 - 55
Schools	
Class rooms up to area 70 Meter Square	35 - 40
Class rooms over 70 Meter Square	30 - 35
Lecture Rooms	30 - 35
Libraries	30 - 40
Hospitals and Clinics	
Private Rooms	25 - 35
Wards, Waiting Rooms	30 - 40
Operating Rooms	25 - 35
Corridor, Laboratory and Public Areas	30 - 40
Audiometric Rooms	20 - 25
Wash Rooms, Toilet, Kitchen	35 - 45
Staff Room, Recreation Room	30 - 40
Ecclesiastical and Academic Buildings	
Church, Mosque	25 - 30
Class Rooms, Lecture Theatre	25 - 35
Laboratory, Workshop	35 - 40
Corridor, Gymnasium	35 - 45

Sound Attenuators

Acoustic Design

NC Vs Sound Pressure Level Table (Table 13a):

NC Level	Sound Pressure Level in dB							
	Octave Band Center Frequency in Hz							
	63	125	250	500	1K	2K	4K	8K
NC 65	80	75	71	68	66	64	63	62
NC 60	77	72	67	63	61	59	58	57
NC 55	74	67	62	58	56	54	53	52
NC 50	71	64	59	54	51	49	48	47
NC 45	67	62	54	49	46	44	43	42
NC 40	64	57	50	45	41	39	38	37
NC 35	60	52	45	40	36	34	33	32
NC 30	59	48	40	34	30	27	25	23
NC 25	54	44	37	31	27	24	22	21
NC 20	51	40	33	26	22	19	17	16

NR Vs Sound Pressure Level Table (Table 13b):

NC Level	Sound Pressure Level in dB							
	Octave Band Center Frequency in Hz							
	63	125	250	500	1K	2K	4K	8K
NR 65	87	79	72	68	65	62	61	59
NR 60	83	74	68	63	60	57	55	54
NR 55	79	70	63	58	55	52	50	49
NR 50	75	65	59	53	50	47	45	43
NR 45	71	61	54	48	45	42	40	38
NR 40	67	57	49	44	40	37	35	33
NR 35	63	52	45	39	35	32	30	28
NR 30	59	48	40	34	30	27	25	23
NR 25	55	44	35	29	25	22	20	18
NR 20	51	39	31	24	20	17	14	13

dBA Vs Sound Pressure Level Table (Table 13c):

NC Level	Sound Pressure Level in dB							
	Octave Band Center Frequency in Hz							
	63	125	250	500	1K	2K	4K	8K
dBA 70	80	75	71	68	66	64	63	62
dBA 65	77	71	67	63	61	59	58	57
dBA 60	74	68	63	58	56	54	53	52
dBA 55	70	64	59	54	51	49	48	47
dBA 50	67	60	55	49	46	44	43	42
dBA 45	64	56	50	45	41	39	38	37
dBA 40	60	52	46	40	36	34	33	32
dBA 35	57	48	42	35	31	29	28	26
dBA 30	54	45	38	31	27	24	24392	21
dBA 25	51	41	33	26	22	19	17	16
dBA 20	47	37	29	22	17	14	12	11

Sound Attenuators

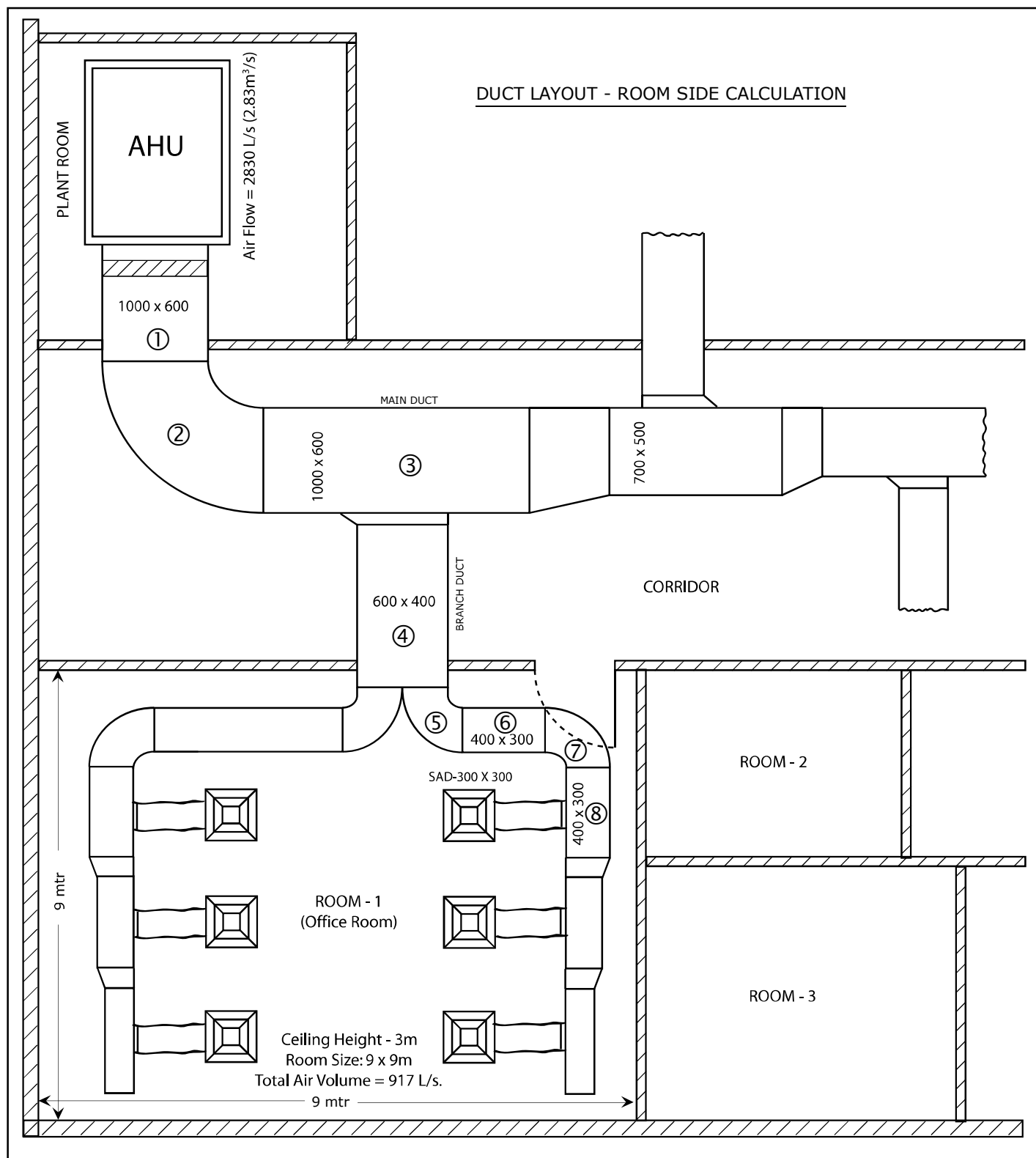
Acoustic Design

Sound Calculation Sheet

				Room Side Calculation							
Project				Date							
Client				Building							
Contractor				Equipment No.							
Engineer				Tag No.							
				Octave Band Centre Frequency (Hz)							
				63	125	250	500	1K	2K	4K	8K
Source Sound Power Level											
Sl.No	Type of Duct	Dimension	Length								
1	Rectangular Duct			-							
2	Rectangular Duct			-							
3	Rectangular Duct			-							
4	Rectangular Duct			-							
5	Rectangular Duct			-							
Sl No	Type of Elbow	Width	Qty								
1	Radiussed Elbows			-							
2	Radiussed Elbows			-							
3	Radiussed Elbows			-							
4	Radiussed Elbows			-							
5	Radiussed Elbows			-							
Sl.No	Additional Attenuation										
1				-							
2				-							
Sl.No		Length (cm)	Width (cm)								
1	Outlet Reflection			-							
SWL Leaving System				-							
Percentage leaving outlet				-							
Distance from outlet to Silencer				-							
Directivity				+							
Direct SPL				-							
Percentage reaching room				-							
Room volume	Length x Width	Height		-							
Reverberation time				+							
Reverberant SPL				-							
Combined SPL											
Criterion NC / NR / dBA											
Safety Factor in dB											
Required Insertion Loss											
Selected Insertion Loss											

Sound Attenuators

Room Side Calculation - Worked Example



Sound Attenuators

Room Side Calculation - Worked Example

Fan Data

Fan Air Volume: 6000cfm=>2830L/s=>2.83m³/s.

Static Pressure: 250Pa.

Fan Sound Power Level:

Octave Band Center Frequency fm in Hz							
63	125	250	500	1K	2K	4K	8K
89	94	93	95	90	88	86	80

Room Details

Room Volume : 243m³

Room height : 3m

Outlet Dimension: 300x300mm

Total No of outlets: 6

Distance from outlet to Listener: 1.5m

Airflow Leaving Outlet: 324cfm=>152L/s=>0.153m³/s.

Total Airflow reaching the room: 1944cfm=>917L/s=>0.920m³/s.

Room type : Office Room

Room Noise Criterion: NC-35.

Ducting Details

S. No	Type	W x H in mm	Length / Remarks
1	Straight Duct	1000 x 600	2.4 Meter
2	Elbow	1000 x 600	Radiussed
3	Straight Duct	1000 x 600	4.6 Meter
4	Straight Duct	600 x 400	6.2 Meter
5	Elbow	400 x 300	Radiussed
6	Straight Duct	400 x 300	1.5 Meter
7	Elbow	400 x 300	Radiussed
8	Straight Duct	400 x 300	2 Meter

Sound Attenuators

Pressure Drop Calculation - Worked Example

Attenuator Pressure Drop Calculation

Determine the face velocity in meter per second from the sound attenuator selected by dividing the air volume in meter cube per second by the attenuator face area in meter square. Plot the face velocity arrived against the attenuator model selected to obtain the attenuator pressure drop.

Example:

Sound attenuator is 1200mm Wide and 700mm High and 1500mm Long, it handles the air volume of 2830L/s. Determine the static pressure drop.

Solution:

$$\begin{aligned}\text{Face area} &= \text{Width} \times \text{Height} \\ &= 1.2 \times 0.7 \\ &= 0.84\text{m}^2\end{aligned}$$

$$\begin{aligned}\text{Face Velocity} &= \text{Air Volume} / \text{Face Area} \\ &= 2.83 / 0.84 \\ &= 3.37\text{m/s}.\end{aligned}$$

From the Aerodynamic Performance data table for Model: CSA-HM/20-100, the attenuator static pressure drop against the face velocity 3.37m/s is 43Pa.

Nomenclature

W - Width in mm

H - Height in mm

L - Length in mm

D - Splitter Thickness in mm.

S-Airway width in mm.

Q-Air Volume in L/s.

V_f -Face Velocity in m/s ($V_f = Q / (W \times H)$).

ΔP -Pressure Loss in Pa.

f_m -Octave Band Centre Frequency in Hz.

D_e -Dynamic Insertion Loss in dB.

NC-Noise Criterion.

Sound Attenuators

Sound Calculation sheet

CENTURY MECHANICAL SYSTEMS FACTORY L.L.C

Sound Calculation Sheet

EXCELAIR
SOUND ATTENUATOR

Project		Worked Example		Date		Room Side Calculation							
Client				Building									
Contractor				Equipment No.									
Engineer				Tag No.									
				Octave Band Centre Frequency (Hz)									
				63	125	250	500	1K	2K	4K	8K		
Source Sound Power Level				89	94	93	95	90	88	86	80		
Sl.No	Type of Duct	Dimension	Length										
1	Rectangular Duct	600	2.4	-	1.44	1.44	0.72	0.36	0.36	0.36	0.36	0.36	
2	Rectangular Duct	600	4.6	-	2.76	2.76	1.38	0.69	0.69	0.69	0.69	0.69	
3	Rectangular Duct	400	6.2	-	3.72	3.72	2.79	1.86	1.24	1.24	1.24	1.24	
4	Rectangular Duct	300	1.5	-	0.90	1.90	0.68	0.45	0.30	0.30	0.30	0.30	
5	Rectangular Duct	300	2	-	1.20	1.20	0.90	0.60	0.40	0.40	0.40	0.40	
Sl.No	Type of Elbow	Width	Qty										
1	Radiussed Elbows	1000	1	-	0	0	1	2	3	3	3	3	
2	Radiussed Elbows	400	1	-	0	0	0	1	2	3	3	3	
3	Radiussed Elbows	400	1	-	0	0	0	1	2	3	3	3	
4	Radiussed Elbows	0	0	-	0	0	0	0	0	0	0	0	
5	Radiussed Elbows	0	0	-	0	0	0	0	0	0	0	0	
Sl.No	Additional Attenuation												
1				-	0	0	0	0	0	0	0	0	
2				-	0	0	0	0	0	0	0	0	
Sl. No		Length (cm)	Width (cm)										
1	Outlet Reflection	30	30	-	12	7	3	1	0	0	0	0	
SWL Leaving System					67	77	83	86	60	76	74	68	
Percentage leaving outlet		0.153 M³/s	5 %	-	13	13	13	13	13	13	13	13	
Distance from outlet to Silencer		1.5 m		-	15	15	15	15	15	15	15	15	
Directivity		Center of Wall or Ceiling		+	4	4	6	7	8	9	9	9	
Direct SPL					43	53	61	65	60	57	55	49	
Percentage reaching room		0.92 M³/s	33 %	-	5	5	5	5	5	5	5	5	
Room volume	Length x Width - 81 m²	Height - 3 m		-	-10	-10	-10	-10	-10	-10	-10	-10	
Reverberation time		1 Second		+	0	0	0	0	0	0	0	0	
Reverberant SPL					52	62	68	71	65	61	59	53	
Combined SPL					53	63	69	72	66	62	60	54	
Criterion NC / NR / dBA		NC 35			60	52	45	40	36	34	33	32	
Safety Factor in dB		2			2	2	2	2	2	2	2	2	
Required Insertion Loss					0	13	26	34	32	30	29	24	
Selected Insertion Loss					11	17	29	45	50	50	42	34	
Sound Attenuator Selection				Octave Band Centre Frequency (Hz)									
CSA-HM/20 - 100 / 1500L X 1200W X 700H				63	125	250	500	1K	2K	4K	8K		
Required Insertion Loss				0	13	26	34	32	30	29	24		
Selected Insertion Loss				11	17	29	45	50	50	42	32		
Attenuator Air Flow (M³/s)		2.83											
Attenuator Width (mm)		1200											
Attenuator Height (mm)		700											
Attenuator Length (mm)		1500											
Attenuator Pressure Drop (Pa)		43											

Sound Attenuators

Ordering Requisites

1.Type

1. Rectangular Attenuator	CSA-H/20-75
2. Rectangular Attenuator	CSA-HM/20-100
3. Rectangular Attenuator	CSA-M/20-125
4. Rectangular Attenuator	CSA-MA/20-150
5. Rectangular Attenuator	CSA-LM/20-175
6. Rectangular Attenuator	CSA-L/20-200
7. Vertical Bend Attenuator	CSA-BV
8. Horizontal Bend Attenuator	CSA-BH
9. Acoustic Splitters	CAS

2. Size

1. Attenuator Width in mm	W
2. Attenuator Height in mm	H
3. Attenuator Length in mm	L

3. Construction

1. Standard Construction	S
2. High Pressure Casing (1.0mm)	HP-1
3. High Pressure Casing (1.2mm)	HP-2
4. Melinex Facing	MF
5. Stainless Steel Construction (Grade 304/2B)	SS-304
6. Stainless Steel Construction (Grade 316/2B)	SS-316
7. Aluminium Construction (Grade 3003)	AL-3003
8. Epoxy Coated	EC
9. Angle Flange (30mm)	AF-30
10. Angle Flange (40mm)	AF-40
11. Angle Flange (50mm)	AF-50

Example

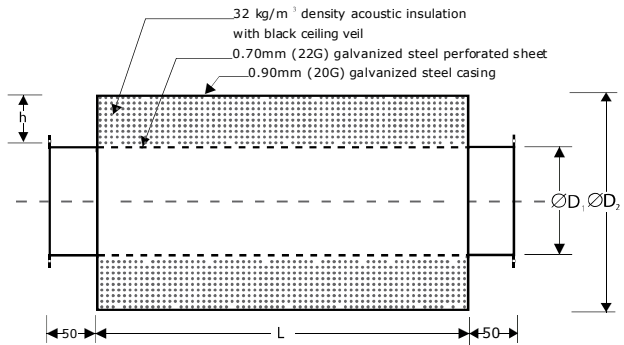
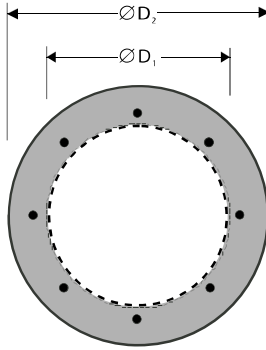
1	2	3
CSA-M/20-125	1200 x 600 x 1500	S, AF-30

Rectangular Sound Attenuator of Model: CSA-M/20-125, 1200 (W) x 600 H) x 1500 mm (L),
Standard GI construction with 30mm Angle Flange.

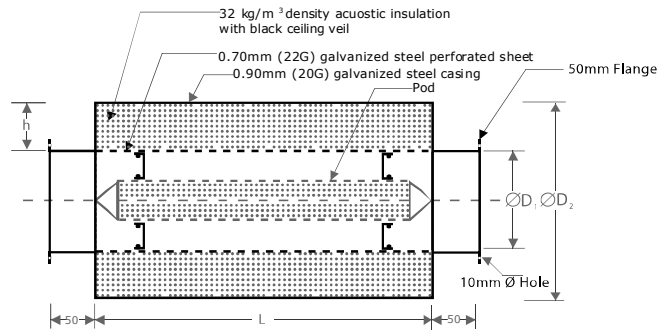
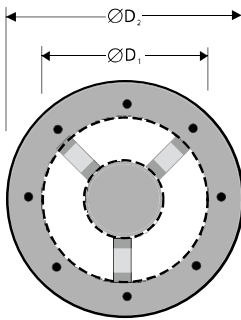
Sound Attenuators

Circular Attenuator

Circular Attenuator without Pod (Model: CSA-CA)



Circular Attenuator with Pod (Model: CSA-CAP)



D_1 - Attenuator Inside (Duct) diameter in mm
 D_2 - Attenuator outside diameter in mm

L - Attenuator length in mm
h - Acoustic Lining thickness in mm

Constructional Details

Circular Sound attenuator casing is constructed from galvanized sheet metal of 0.90mm (20G) thickness. Casings thickness confirms to SMACNA & DW 144 class B standards. Both ends are provided with flange which contain holes to connect with the duct work.

Sound Attenuators

Constructional Detail

The standard 100mm thickness acoustic media (lining) is covered with 0.70mm (22G) thickness of galvanized steel perforated sheets. Acoustic lining is filled with not less than 32kg/m³ density class 'O' building regulated, inorganic semi rigid Rockwool fiber slab (mineral wool) acoustic filler material. This filler material is inert, non-hygroscopic, vermin & rots proof, does not support bacterial & fungus growth. The acoustic filler has a black ceiling veil (black facing) and packed with galvanized steel perforated sheet, this dual protection prevents damage and fiber erosion up to 30m/s air way velocity.

The sound absorbing pod fitted (Model: CSA-CAP) in the centre has galvanized perforated sheet covering and coned at both ends to minimize the pressure loss and regenerated noise from the attenuator. Pod is secured to the attenuator casing by means of rivets.

Sound attenuator with melinex (Polyester membrane) facing is available as option. Polyester membrane is used to protect the acoustic lining in oil or grease or in clean or sterile or hospital applications.

The acoustic infill is complying to following standards,

Fire resistance	: ASTM E 119 & E84, ASTM C447 & BS 476.
Acoustic Performance	: ASTM C423, ASTM E413 & E90 & BS 3638/ISO 354.
Moisture Absorption	: ASTM C209/14, BS 2972/12, ASTM C1104
Corrosion Resistance	: ASTM C692.

CSA//HP: Sound attenuator Model: CSA-CA/HP & CSA-CAP/HP are suitable for high pressure application with increased casing thickness to comply the DW144 class C or D and SMACNA standard (high pressure).

Sound Attenuators

Constructional Detail

CSA//MF: Sound attenuator Model: CSA-CA/MF & CSA-CAP/MF are normal circular sound attenuator with the acoustic infill is faced with Melinex (Polyester membrane) which is suitable for hospital or oil or grease applications.

Attenuators are supplied in sections when the Length of the attenuator is exceeded : Length=2000mm. Each attenuator can be divided into equal sizes to achieve the total length.

The assembly of sectionalized sound attenuators at site to be carried out by the client. The full details of assembly and configuration will be provided by CMS. Coupler will be provided by us.

Optional Constructional Specification:

Galvanized steel construction	: 1.0 mm GI casing (Model: CSA//HP-1) 1.2mm GI casing (Model: CSA//HP-2)
Stainless Steel Construction	: 1.0 mm thickness, Grade 304 with 2B mill finish (Model: CSA//SS-304). 1.0 mm thickness, Grade 316 with 2B mill finish (Model: CSA//SS-316).
Aluminium Construction	: 1.0mm thickness Grade 3003 H14 (Model: CSA//AL-3003)
Additional Coating	: Epoxy coated (Model: CSA//EC)
Optional Acoustic Filler	: Fiber Glass (Model: CSA//FG)
Additional Facing	: Melinex Facing (Model: CSA//MF)

Nomenclature:

	V_f -Face velocity in m/s ($V_f=Q/D_1$).
D_1 -Attenuator inside duct diameter in mm	ΔP -Pressure Loss in Pa.
D_2 -Attenuator outside duct diameter in mm	f_m -Octave Band Centre Frequency in Hz.
L-Attenuator Length in mm.	D_e -Dynamic Insertion Loss in dB.
h-Lining thickness in mm	NC-Noise Criterion.
Q-Air Volume in L/s.	

Sound Attenuators

Aerodynamic Performance Data

Model: CSA-CAP/1D

Attenuator Size (Ø mm)		300	400	500	600	700	800	900	1000	1100	1200
Face Area A (m ²)		0.071	0.126	0.196	0.283	0.385	0.502	0.636	0.785	0.950	1.130
Pressure Drop ΔP (Pa)	Face Velocity V _f (m/s)	Air Volume Q (L/s)									
10	6.2	440	781	1215	1755	2387	3112	3943	4867	5890	7006
15	7.1	504	895	1392	2009	2734	3564	4516	5574	6745	8023
20	8.2	582	1033	1607	2321	3157	4116	5215	6437	7790	9266
25	9.0	639	1134	1764	2547	3465	4518	5724	7065	8550	10170
30	10.2	724	1285	1999	2887	3927	5120	6487	8007	9690	11526
35	11.0	781	1386	2156	3113	4235	5522	6996	8635	10450	12430
40	12.0	852	1512	2352	3396	4620	6024	7632	9420	11400	13560
45	12.5	888	1575	2450	3538	4813	6275	7950	9813	11875	14125
50	13.2	937	1663	2387	3736	5082	6626	8395	10362	12540	14916
60	14.5	1030	1827	2842	4104	5583	7279	9222	11383	13775	16385
70	15.2	1079	1915	2979	4302	5852	7630	9667	11932	14440	17176
80	16.0	1136	2016	3136	4528	6160	8032	10176	12560	15200	18080
90	16.8	1193	2117	3293	4754	6468	8434	10685	13188	15960	18984
100	18.0	1278	2268	3528	5094	6930	9036	11448	14130	17100	20340
125	19.4	1377	2444	3802	5490	7469	9739	12338	15229	18430	21922
150	21.0	1491	2646	4116	5943	8085	10542	13356	16485	19950	23730
200	25.5	1811	3213	4998	7217	9818	12801	16218	20018	24225	28815

Above pressure loss data is related to a straight pod type attenuators at airflow to the attenuator is uniform over the face in a duct system. Podless attenuators will have pressure loss similar to an equivalent length of straight duct work, hence excluded from the table.

Sound Attenuators

Aerodynamic Performance Data

Model: CSA-CAP/2D

Attenuator Size (Ø mm)		300	400	500	600	700	800	900	1000	1100	1200
Face Area A (m ²)		0.071	0.126	0.196	0.283	0.385	0.502	0.636	0.785	0.950	1.130
Pressure Drop ΔP (Pa)	Face Velocity V_f (m/s)	Air Volume Q (L/s)									
15	4.6	327	580	902	1302	1771	2309	2926	3611	4370	5198
20	6.2	440	781	1215	1755	2387	3112	3943	4867	5890	7006
25	7.4	525	932	1450	2094	2849	3715	4706	5809	7030	8362
30	8.1	575	1021	1588	2292	3119	4066	5152	6359	7695	9153
35	8.6	611	1084	1686	2434	3311	4317	5470	6751	8170	9718
40	9.0	639	1134	1764	2547	3465	4518	5724	7065	8550	10170
45	9.9	703	1247	1940	2802	3812	4970	6296	7772	9405	11187
50	10.8	767	1361	2117	3056	4158	5422	6869	8478	10260	12204
60	11.8	838	1487	2313	3339	4543	5924	7505	9263	11210	13334
70	13.2	937	1663	2587	3736	5082	6626	8395	10362	12540	14916
80	14.5	1030	1827	2842	4104	5583	7279	9222	11383	13775	16385
90	15.0	1065	1890	2940	4245	5775	7530	9540	11775	14250	16950
100	16.0	1136	2016	3136	4528	6160	8032	10176	12560	15200	18080
125	16.8	1193	2117	3293	4754	6468	8434	10685	13188	15960	18984
150	18.2	1292	2293	3567	5151	7007	9136	11575	14287	17290	20566
200	21.0	1491	2646	4116	5943	8085	10542	13356	16485	19950	23730

Above Pressure loss data is related to a straight pod type attenuators at airflow to the attenuator is uniform over the face in a duct system. Podless attenuators will have pressure loss similar to an equivalent length of duct work, hence excluded from the table.

Sound Attenuators

Dynamic Insertion Loss

Model: CSA-CA (without Pod)

Attenuator Size (Ø mm)	Attenuator Model	Length (mm)	Octave Band Center Frequency (Hz)							
			63	125	250	500	1K	2K	4K	8K
250	CSA-CA/1D/250	250	2	2	4	7	11	10	8	7
	CSA-CA/2D/250	500	2	2	5	9	12	11	8	8
300	CSA-CA/1D/300	300	2	3	6	8	13	11	9	7
	CSA-CA/2D/300	600	2	3	6	12	18	13	10	9
400	CSA-CA/1D/400	400	2	4	6	10	14	12	9	8
	CSA-CA/2D/400	800	3	5	8	14	19	15	11	10
500	CSA-CA/1D/500	500	2	5	6	12	15	12	10	7
	CSA-CA/2D/500	1000	3	7	9	18	24	22	18	12
600	CSA-CA/1D/600	600	3	6	8	14	16	12	8	7
	CSA-CA/2D/600	1200	4	8	12	21	25	24	20	15
700	CSA-CA/1D/700	700	3	7	9	16	18	12	8	7
	CSA-CA/2D/700	1400	4	9	16	22	26	22	19	14
800	CSA-CA/1D/800	800	3	8	10	18	19	12	10	8
	CSA-CA/2D/800	1600	4	10	18	23	26	20	20	15
900	CSA-CA/1D/900	900	4	8	11	18	20	11	10	9
	CSA-CA/2D/900	1800	5	9	18	24	24	19	17	13
1000	CSA-CA/1D/1000	1000	5	8	12	19	21	12	10	10
	CSA-CA/2D/1000	2000	6	10	18	25	25	18	15	12
1100	CSA-CA/1D/1100	1100	6	8	13	20	20	11	9	9
	CSA-CA/2D/1100	2200	7	11	20	26	27	19	13	10
1200	CSA-CA/1D/1200	1200	6	9	14	21	18	10	10	8
	CSA-CA/2D/1200	2400	7	12	22	28	27	20	15	11

Model: CSA-CAP (with Pod)

Attenuator Size (Ø mm)	Attenuator Model	Length (mm)	Octave Band Center Frequency (Hz)							
			63	125	250	500	1K	2K	4K	8K
300	CSA-CA/1D/300	300	2	7	9	12	20	22	19	18
	CSA-CA/2D/300	600	4	10	14	20	26	28	28	21
400	CSA-CA/1D/400	400	2	8	10	15	22	25	21	19
	CSA-CA/2D/400	800	4	10	16	24	29	30	29	22
500	CSA-CA/1D/500	500	3	8	11	18	23	26	22	17
	CSA-CA/2D/500	1000	4	11	18	29	30	31	30	21
600	CSA-CA/1D/600	600	3	8	12	20	27	25	22	18
	CSA-CA/2D/600	1200	5	12	20	28	30	32	28	20
700	CSA-CA/1D/700	700	4	8	12	21	26	24	20	17
	CSA-CA/2D/700	1400	6	12	21	29	31	32	27	22
800	CSA-CA/1D/800	800	5	8	12	22	28	25	21	16
	CSA-CA/2D/800	1600	6	11	19	30	32	32	28	21
900	CSA-CA/1D/900	900	5	9	13	23	27	22	18	14
	CSA-CA/2D/900	1800	7	12	20	30	30	29	27	20
1000	CSA-CA/1D/1000	1000	6	9	14	24	26	23	19	16
	CSA-CA/2D/1000	2000	8	13	21	31	30	29	26	19
1100	CSA-CA/1D/1100	1100	6	9	15	26	25	22	17	12
	CSA-CA/2D/1100	2200	8	12	22	31	29	27	22	17
1200	CSA-CA/1D/1200	1200	7	9	17	27	24	21	18	14
	CSA-CA/2D/1200	2400	9	14	22	32	30	28	24	16

* Above performance data applies when both noise and air flow are travelling in the same direction.

Insertion Loss Correction table for melinex acoustic infill:

Multiplication Factor	x 0.98	x 0.79	x 0.81	x 0.92	x 0.94	x 0.68	x 0.60	x 0.78
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Sound Attenuators

Self Generated Noise

Model: CSA-CAP

Attenuator Length (mm)	Face Velocity (m/s)	Self Generated Noise (dB)							
		Octave Band Center Frequency (Hz)							
		63	125	250	500	1k	2k	4k	8k
300	5.0	36	30	29	24	19	15	<20	<20
	7.5	41	38	34	31	27	24	<20	<20
	10.0	49	45	41	40	36	34	28	<20
	15.0	57	54	51	50	48	45	38	30
	20.0	64	62	56	59	58	54	46	38
400	5.0	38	32	31	26	21	17	<20	<20
	7.5	43	40	36	33	29	26	22	<20
	10.0	51	47	43	42	38	36	30	21
	15.0	59	56	53	52	50	47	40	32
	20.0	66	64	58	61	60	56	48	40
500	5.0	39	33	32	27	22	18	<20	<20
	7.5	44	41	37	34	30	27	23	<20
	10.0	52	48	44	43	39	37	31	22
	15.0	60	57	54	53	51	48	41	33
	20.0	67	65	59	62	61	57	49	41
600	5.0	41	35	34	29	24	20	<20	<20
	7.5	46	43	39	36	32	29	25	<20
	10.0	54	50	46	45	41	39	33	24
	15.0	62	59	56	55	53	50	43	35
	20.0	69	67	61	64	63	59	51	43
700	5.0	42	36	35	30	25	21	<20	<20
	7.5	47	44	40	37	33	30	26	<20
	10.0	55	51	47	46	42	40	34	25
	15.0	64	61	58	57	55	52	45	37
	20.0	71	69	63	66	65	61	53	45
800	5.0	43	37	36	31	26	22	<20	<20
	7.5	48	45	41	38	34	31	27	23
	10.0	56	52	48	47	43	41	35	26
	15.0	64	61	58	57	55	52	45	37
	20.0	71	69	63	66	65	61	53	45
900	5.0	43	37	36	31	26	22	<20	<20
	7.5	48	45	41	38	34	31	27	23
	10.0	57	53	49	48	44	42	36	27
	15.0	65	62	59	58	56	53	46	38
	20.0	72	70	64	67	66	62	54	46
1000	5.0	45	39	38	33	28	24	<20	<20
	7.5	50	47	43	40	36	33	29	23
	10.0	58	54	50	49	45	43	37	28
	15.0	66	63	60	59	57	54	47	39
	20.0	73	71	65	68	67	63	55	47
1100	5.0	46	40	39	34	29	25	<20	<20
	7.5	51	48	44	41	37	34	30	24
	10.0	59	55	51	50	46	44	38	29
	15.0	67	64	61	60	57	54	49	41
	20.0	74	71	66	70	68	64	57	48
1200	5.0	47	41	40	35	30	26	<20	<20
	7.5	52	49	45	42	38	35	31	25
	10.0	60	56	52	51	47	45	39	30
	15.0	68	65	62	61	59	56	49	41
	20.0	75	73	67	70	69	65	57	49

* Above self generated noise data applies when both noise and air flow are travelling in the same direction. Air generated noise in sound attenuator without pod is equivalent to the value of circular duct.

Sound Attenuators

Dimension & Weight Details

Dimensional Details

Attenuator Size (mm)	D ₁ (mm)	D ₂ (mm)	h (mm)	Length, L=1xD ₁ (mm)	Length, L=2xD ₁ (mm)	d (mm)
250	250	455	100	250	500	10
300	300	505	100	300	600	10
400	400	605	100	400	800	10
500	500	705	100	500	1000	10
600	600	805	100	600	1200	10
700	700	905	100	700	1400	10
800	800	1005	100	800	1600	10
900	900	1105	100	900	1800	10
1000	1000	1205	100	1000	2000	10
1200	1200	1405	100	1200	2400	10

D₁ - Attenuator Inside (Duct) Diameter in mm.

D₂ - Attenuator Outside Diameter in mm.

h - Acoustic Lining thickness in mm

L - Attenuator Length in mm

d - Flange hole Diameter in mm

Weight Details

Attenuator Size (mm)	Attenuator Weight in kg (+/- 10%)			
	CSA-CA/1D	CSA-CA/2D	CSA-CAP/1D	CSA-CAP/2D
250	8	11	--	--
300	10	14	13	18
400	16	24	23	32
500	26	37	36	51
600	36	50	49	72
700	45	64	63	90
800	52	73	71	99
900	63	90	87	120
1000	85	120	108	165
1200	120	160	150	210

Sound Attenuators

Ordering Requisites

1.Type

- | | |
|------------------------------------|---------|
| 1. Circular Attenuator without pod | CSA-CA |
| 2. Circular Attenuator with pod | CSA-CAP |

2. Size

- | | |
|------------------------------|---|
| 1. Attenuator Diameter in mm | D |
| 2. Attenuator Length in mm | L |

3. Construction

- | | |
|--|---------|
| 1. Standard Construction | S |
| 2. High Pressure Casing (1.0mm) | HP-1 |
| 3. High Pressure Casing (1.2mm) | HP-2 |
| 4. Melinex Facing | MF |
| 5. Stainless Steel Construction (Grade 304/2B) | SS-304 |
| 6. Stainless Steel Construction (Grade 316/2B) | SS-316 |
| 7. Aluminium Construction (Grade 3003) | AL-3003 |
| 8. Epoxy Coated | EC |

Example

1	2	3
CSA-CAP	1000 Ø x 1200 L	S, EC

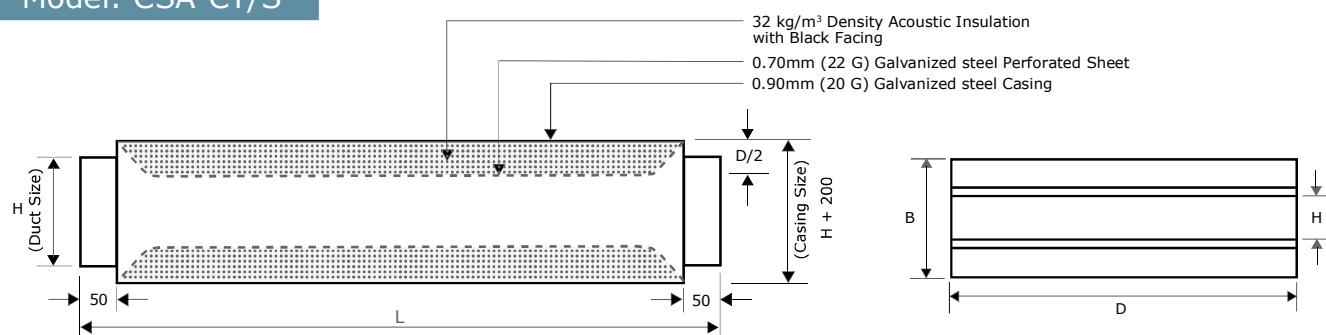
Circular Sound Attenuator of Model: CSA-CAP, 1000 Ø x 1200 mm (L), Standard GI construction with epoxy Coating.

Sound Attenuators

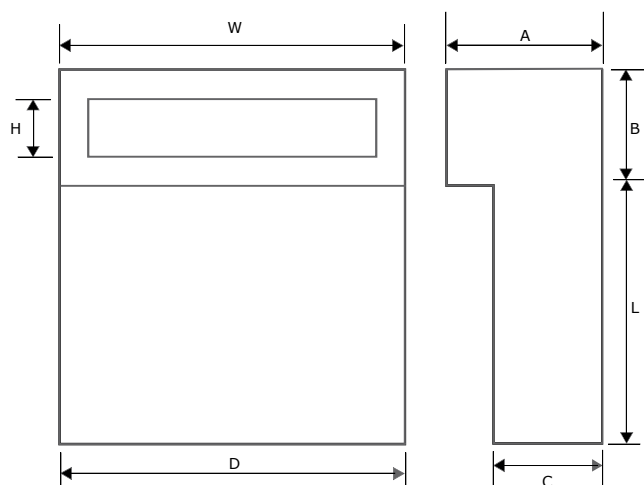
Crosstalk Attenuator

- Installed where privacy and security have priority while maintaining a continuous flow of air.
- Maintains maximum noise suppressions and minimum noise interference between inter connected rooms.
- Fulfills the necessary requirement of "room to room" cross talk application where no duct is required or used.
- Crosstalk attenuators are designed to avoid noise radiated from a room to room or from a duct termination which is connected through a main duct to a branch duct serving to another room.

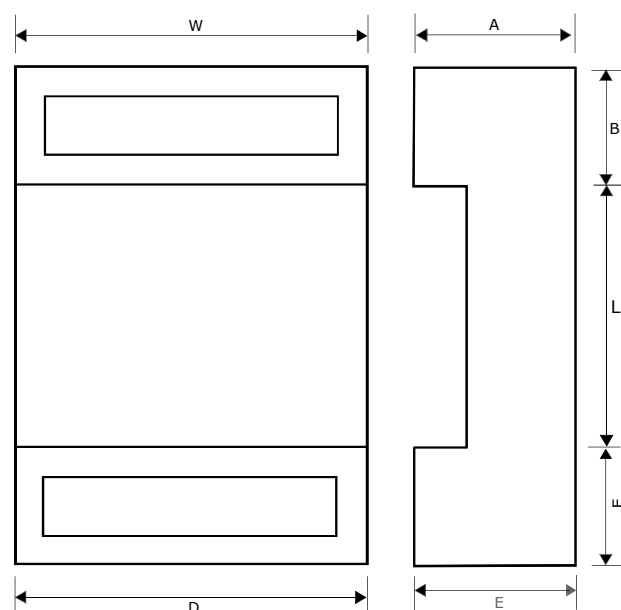
Model: CSA-CT/S

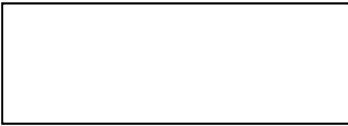


Model: CSA-CT/L



Model: CSA-CT/U





Optional Egg Crate Grille

Sound Attenuators

Types of CT Attenuator

S-Type (Model: CSA-CT/S)

This type of attenuator is designed for in-line duct mounting in an air-conditioning system where rooms are served by branches of a common duct system or suitable for room to room attenuation. Splitters are radiussed at both ends to achieve the minimum pressure drop and self-generated noises. It can be supplied with optional grilles.

L-type (Model: CSA-CT/L)

L type cross talk attenuators are designed to avoid noise interference between interconnected rooms. They are mounted between the wall and ceiling. It is used as air transfer attenuator between adjoining areas where the acoustic integrity of a common partition or room or ceiling needs to be maintained. It can be supplied with optional grilles.

U-Type (Model: CSA-CT/U)

U type cross talk attenuators are designed to avoid noise interference between interconnected rooms. They are mounted between the two ceilings. It is used as air transfer attenuator between adjoining areas where the acoustic integrity of a common partition or room or ceiling needs to be maintained. It can be supplied with optional grilles.

Z-Type (Model: CSA-CT/Z)

Z type cross talk attenuators are designed to avoid noise interference between interconnected rooms. They are mounted within the wall. It is used as air transfer attenuator between adjoining areas where the acoustic integrity of a common partition or room or ceiling needs to be maintained. It can be supplied with optional grilles.

Advantages

- Maintains high privacy and security at all air flow conditions.
- Effective for maximum speech and noise suppression.
- Radius splitter construction offers minimum pressure drop and regenerated noise level.
- Compatible to install and adjust.

Installation

- Can be installed in walls or partitions or doors or suspended ceilings.
- Can be installed inline with duct work.
- Tight air seal to be maintained between the unit and the wall / ceiling opening.

Sound Attenuators

Dimension Details

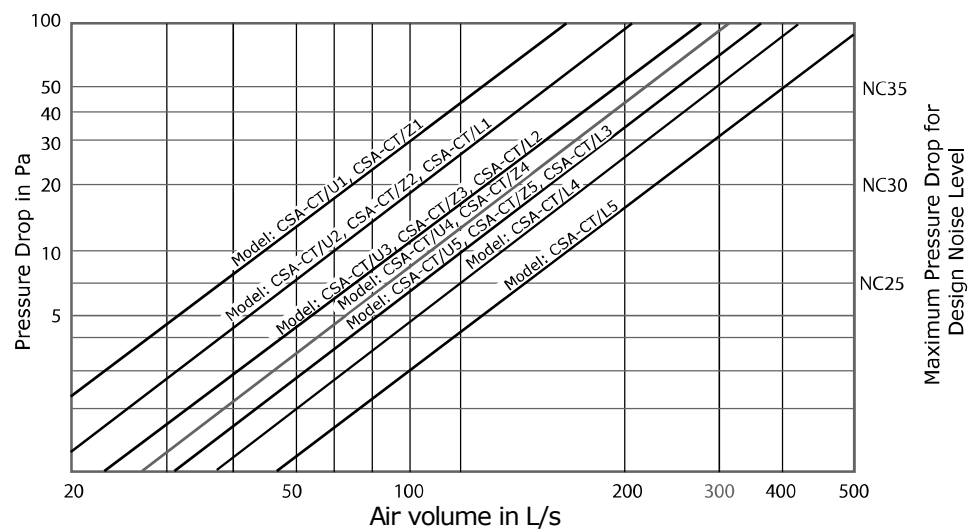
S. No	W (mm)	H (mm)	L (mm)	A (mm)	B (mm)	C (mm)	D (mm)	E (mm)	F (mm)
Model: CSA-CT/S									
1	500	100	500	-	300	-	-	-	-
2	700	100	750	-	300	-	-	-	-
3	900	100	1000	-	300	-	-	-	-
4	1000	100	1250	-	300	-	-	-	-
5	1200	100	1500	-	300	-	-	-	-
Model: CSA-CT/L									
1	500	100	500	250	200	200	500	-	-
2	700	100	750	250	200	200	700	-	-
3	900	100	1000	250	200	200	900	-	-
4	1000	100	1250	250	200	200	1000	-	-
5	1200	100	1500	175	200	200	1200	-	-
Model: CSA-CT/U									
1	500	100	500	250	200	200	500	250	-
2	700	100	750	250	200	200	700	250	-
3	900	100	1000	250	200	200	900	250	-
4	1000	100	1250	250	200	200	1000	250	-
5	1200	100	1500	250	200	200	1200	250	-
Model: CSA-CT/Z									
1	500	100	500	175	300	300	500	250	200
2	700	100	750	175	300	300	700	250	200
3	900	100	1000	175	300	300	900	250	200
4	1000	100	1250	175	300	300	1000	250	200
5	1200	100	1500	175	300	300	1200	250	200

* All the above models can be supplied with optional Egg crate / Normal Grille at both the inlet & outlet

Sound Attenuators

Pressure Loss chart

Pressure Loss Chart



Dynamic Insertion Loss D_e in dB (Model: CSA-CT/S)

Attenuator Length (mm)	Octave Band Center Frequency fm in Hz							
	63	125	250	500	1K	2K	4K	8K
500	4	6	8	14	21	15	11	9
750	5	8	13	20	31	26	19	14
1000	7	10	17	28	44	34	26	18
1250	10	15	24	25	50	45	33	27
1500	11	16	28	44	50	50	36	30

Attenuator Length against Noise Reduction (Model: CSA-CT/S)

Design Noise Level (NC)	Attenuator Length (mm)	Noise Level at 500Hz (dB)
NC 45	500	30
NC 40	750	35
NC 35	1000	40
NC 30	1250	45
NC 25	1500	50

Sound Attenuators

Performance Data

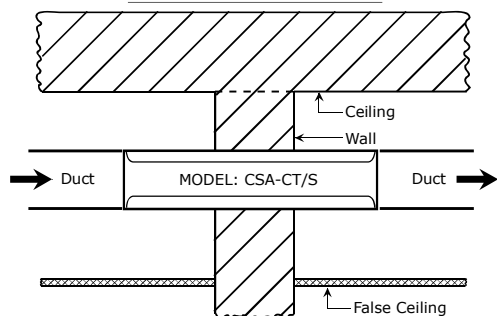
Model: CSA-CT/S

Attenuator Model	Face Velocity (m/s)			2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5
	Self Generated Noise Level			NC 25		NC 30		NC 35		NC 40	
	Attenuator Dimension			Air Volume Q (L/s)							
	Width (mm)	Height (mm)	Length (mm)								
CSA-CT/S-500	100	100	500	20	25	30	35	40	45	50	55
	150	100		30	38	45	53	60	68	75	83
	200	100		40	50	60	70	80	90	100	110
Pressure Drop (Pa)				<10	<10	<10	<10	<10	<10	<10	10
CSA-CT/S-750	150	150	750	45	56	68	79	90	101	113	124
	200	150		60	75	90	105	120	135	150	165
	250	150		75	94	113	131	150	169	188	206
	300	150		90	113	135	158	180	203	225	248
Pressure Drop (Pa)				<10	<10	<10	<10	<10	<10	10	12
CSA-CT/S-1000	200	200	1000	80	100	120	140	160	180	200	220
	250	200		100	125	150	175	200	225	250	275
	300	200		120	150	180	210	240	270	300	330
	350	200		140	175	210	245	280	315	350	385
	400	200		160	200	240	280	320	360	400	440
Pressure Drop (Pa)				<10	<10	<10	<10	12	16	22	26
CSA-CT/S-1250	250	250	1250	125	156	188	219	250	281	313	344
	300	250		150	188	225	263	300	338	375	413
	350	250		175	219	263	306	350	394	438	481
	400	250		200	250	300	350	400	450	500	550
	450	250		225	281	338	394	450	506	563	619
	500	250		250	313	375	438	500	563	625	688
Pressure Drop (Pa)				8	12	18	24	30	38	45	55
CSA-CT/S-1500	300	300	1500	180	225	270	315	360	405	450	495
	350	300		210	263	315	368	420	473	525	578
	400	300		240	300	360	420	480	540	600	660
	450	300		270	338	405	473	540	608	675	743
	500	300		300	375	450	525	600	675	750	825
	550	300		330	413	495	578	660	743	825	908
	600	300		360	450	540	630	720	810	900	990
Pressure Drop (Pa)				14	20	28	36	50	60	80	100

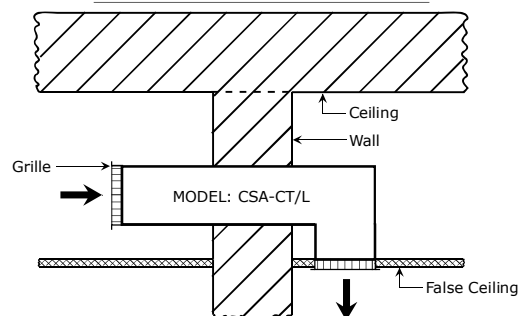
Sound Attenuators

Mounting Arrangements

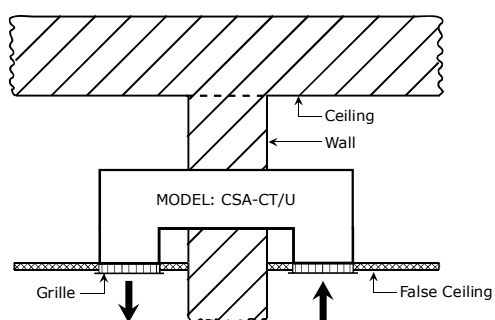
Wall / Duct Mounted



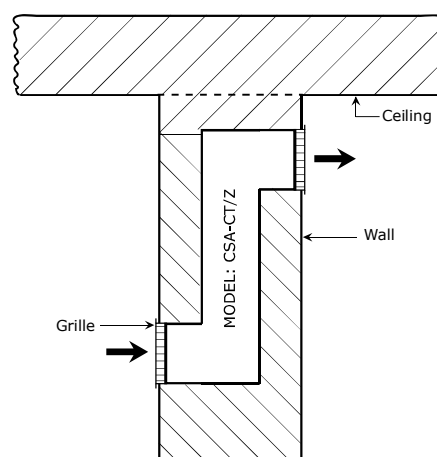
Concealed Ceiling Mounted



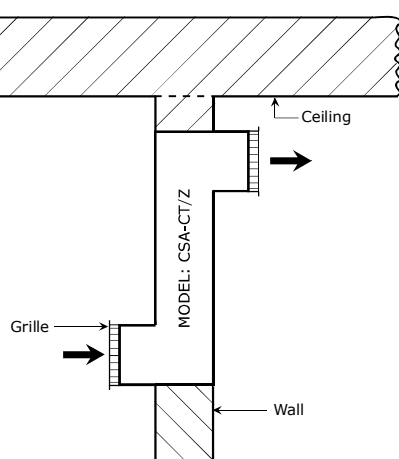
Ceiling Mounted



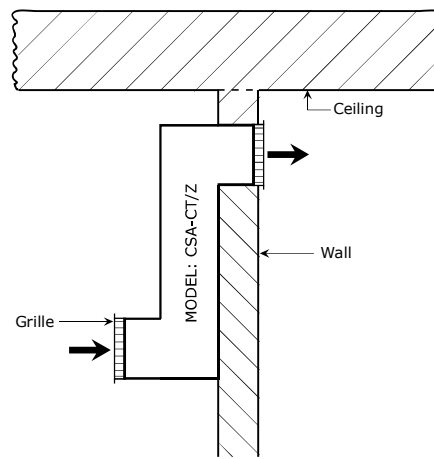
Concealed Wall Mounted



Wall - Recess Mounted



Wall - Surface Mounted



Sound Attenuators

Ordering Requisites

1.Type

- | | |
|-----------------------------------|----------|
| 1. Cross talk attenuator 'S' type | CSA-CT/S |
| 2. Cross talk attenuator 'L' type | CSA-CT/L |
| 3. Cross talk attenuator 'U' type | CSA-CT/U |
| 4. Cross talk attenuator 'Z' type | CSA-CT/Z |

2. Size

- | | |
|----------------------------|---|
| 1. Attenuator Width in mm | W |
| 2. Attenuator Height in mm | H |
| 3. Attenuator Length in mm | L |

3. Construction

- | | |
|--|---------|
| 1. Standard Construction | S |
| 2. Melinex Facing | MF |
| 3. Stainless Steel Construction (Grade 304/2B) | SS-304 |
| 4. Stainless Steel Construction (Grade 316/2B) | SS-316 |
| 5. Aluminium Construction (Grade 3003) | AL-3003 |
| 6. Epoxy Coated | EC |

Example

1	2	3
CSA-CT/U	900 X 100 X 1000	S, EC

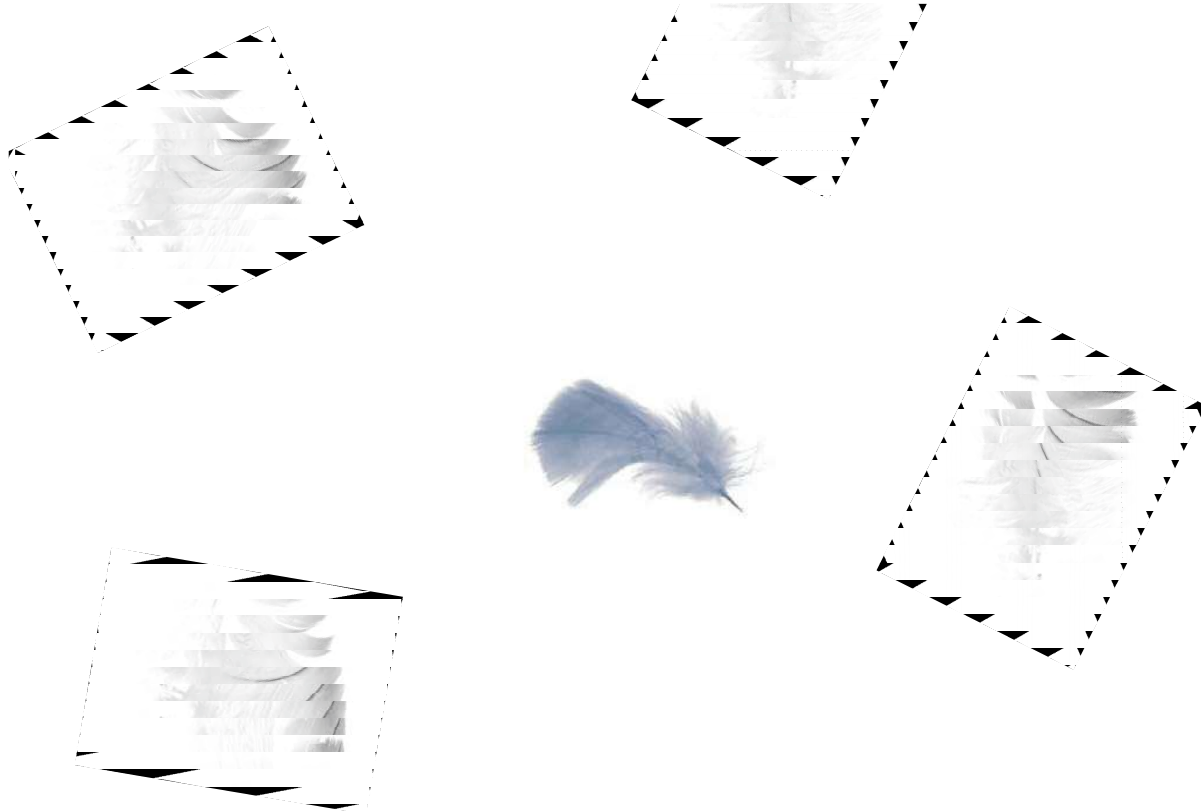
Cross Talk Sound Attenuator of Model: CSA-CT/U, 900 () x 100 (H) x 1000MM (L), Standard GI construction with epoxy Coating.

EXCELAIR GD1000 SERIES

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